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SUPERSEDING
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MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, DIESEL-ENGINE-DRIVEN: PNEUMATIC-TIRED, ROUGH TERRAIN; 4,000 POUND CAPACITY AT 24-INCH LOAD CENTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for diesel-engine-driven, four-wheel drive, rough terrain, pneumatic tired forklift trucks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photo-sensitive Anodized Aluminum).
QQ-P-35	- Passivation Treatments for Corrosion Resisting Steels.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3930

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ZZ-H-428	- Hose, Rubber: Preformed and Straight: Hose, Preformed: Flexible, Wire-Reinforced, (for the Coolant Systems of Automotive and Other Liquid-Cooled Engines).
ZZ-R-765	- Rubber, Silicone: Low- and High-Temperature and Tear Resistant.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-H-3177	- Heaters, Coolant, Engine: Multifuel-Burning, 24-Volt DC.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-R-6130	- Rubber, Cellular, Chemically Blown.
MIL-H-7365	- Hose, Air Duct, for Ground Heaters.
MIL-A-8421	- Air Transportability Requirements.
MIL-H-8796	- Hose, Airduct, Flexible, Aircraft.
MIL-H-13444	- Hose and Hose Assemblies, Rubber: Fuel and Oil.
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-C-21567	- Compound, Silicone, Soft Film.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Glove Set, Chemical Protective.
MIL-F-43987	- Footwear Cover, Chemical Protective (Over-boots).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-H-46792	- Heater, Vehicular Compartment: Multifuel Burning, 24 Volt, DC.
MIL-M-51282	- Mask, Chemical - Biological, Field, M17A1.
MIL-H-51291	- Hood, Chemical - Biological Mask: M6A2.

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MIL-A-52363	- Air Cleaners, Intake: Dry-Type (for Internal-Combustion Engine).
MIL-H-52471	- Hose and Hose Assemblies, Rubber: Hydraulic Pressure-Type, General Specification for.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.
MIL-R-52899	- Ramp Loading Vehicle.
MIL-H-62028	- Hose, Air Duct: High Temperature, Flexible, Reinforced.
MIL-T-62314	- Test Equipment (Simplified) for Internal Combustion Engines (STE/ICE).

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1223	- Administrative Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warrenty Notice Standards.
MIL-STD-1366	- Material Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS51113	- Switch, Vehicular Lights: 24 Volt DC; (Waterproof).

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MS51318	- Headlight: Blackout, 24 Volt, Waterproof.
MS51330	- Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51335	- Pintle Assembly, Towing, 18,000 lbs. Capacity, Manual Release.
MS52125	- Composite Light - Tail, Stop, Turn and Marker.
MS52126	- Composite Light - Front, Turn, Park and Marker.
MS52131	- Connector, Plug, Electrical Intervehicle Power Cable.
MS53052	- Bracket Assembly, Liquid Container, Five Gallon.

MILITARY HANDBOOK

Air Force System Command

Design Handbook DH1-11	- Air Transportability.
MIL-HDBK-157	- Transportability Criteria.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

TACOM DOCUMENTS AND DRAWINGS

Report No. CR-82-588-003	- STE/ICE Design Guide for Vehicle Diagnostic Connector Assemblies.

DWG 12258941 - Connector, Receptacle 54 Pin.
DWG 12258955 - Vehicle Test Card.

(Application for copies should be addressed to the Department of the Army, U.S. Army Tank-Automotive Command, ATTN: DRSTA-RGD, Warren, MI 48090.)

UNITED STATES DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Bulletin No. 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Avenue, Ridge Park, IL 60068.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B56.1	Safety Standards for Low Lift and High Lift Trucks.
B56.6	Safety Standard for Rough Terrain Fork Lift Trucks.
B93.30M	Hydraulic Fluid Power - Contamination Analysis Data - Reporting Method.
B93.42	Testing Hydraulic Fluid Power Quick Disconnect Couplings.
MH5.1.1M	Requirements for Closed Van Cargo Containers.
MH11.3	Load Handling Symbols for Powered Industrial Trucks.
Z26.1	Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1171 Rubber Deterioration - Surface Ozone Cracking Outdoors or Chamber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.
D14.3 - Earthmoving and Construction Equipment Welding, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 North LeJeune Road, Miami, FL 33126.)

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ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 50 East Van Buren Street, IL 60605.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Standard 114 - Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL FLUID POWER ASSOCIATION

T.3.9.18 - 1976 - NFPA Recommended Standard for Calibration of Liquid Automatic Particle Counters.

(Application for copies should be addressed to the National Fluid Power Association, P.O. Box 49, Thiensville, WI 53092.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The forklift truck (hereinafter called "truck" or "forklift") shall be a four-wheel drive, pneumatic tired, diesel-engine-driven, front loading truck.

3.1.1 Rated load. Rated load for the forklift shall be 4,000 pounds measured with the load center (see 6.9.1) of the weight 24 inches from the forward face of the vertical member of the tines and 24 inches above the load-bearing surface of the forks (see 6.9.2).

3.1.2 Weight and dimensions.

3.1.2.1 Weight. The weight of the truck shall not exceed 11,000 pounds and shall apply with the forklift fully fueled, serviced, and ready for operation, without operator.

3.1.2.2 Overall height. The overall height of the forklift including ROPS and FOPS, shall not exceed 80 inches. Height shall be measured as defined in SAE J1234.

3.1.2.3 Length. The length of the forklift, not including forks, shall not exceed 166 inches, and shall be measured as defined in J1234.

3.1.2.4 Width. The width of the forklift, including tires, shall not exceed 79 inches, and shall be measured in accordance with SAE J894.

3.1.2.5 Underclearance. The underclearance of the forklift, including mast, shall not be less than 9 inches and shall apply with the forklift fully fueled and serviced, with rated load in carry position (see 6.9.3). The forklift, with and without load, shall be capable of negotiating a 20-degree ramp with level surfaces at both top and bottom of the ramp. No portion of the forklift, other than the tires, shall come in contact with the ramp when the forklift travels over the ramp in either direction. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by removable structural plates from striking obstacles.

3.1.3 Transportability. The forklift shall be transportable worldwide without damage, by highway, air, marine, and rail transport mode. Transportability criteria are established by MIL-STD-1366 and MIL-HDBK 157 as described herein.

3.1.3.1 Highway transportability. The forklift shall be capable of being towed with a dead engine on improved roads at speeds of at least 35 mph, for not less than 50 miles, without damage. When towed, the forklift shall track the towing vehicle within the width of the forklift tire. A towbar shall be furnished and shall be of an A-frame or V-frame design, hinged at two points on the forklift, and with a solid towbar eye in accordance with SAE J847. The towbar shall provide for towing by trucks with pintle locations in accordance

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with SAE J849, and shall provide for a turning clearance area in accordance with figure 1. A lockup latch shall be provided to secure the towbar in a vertical position when not in use. Two safety chains, equipped with safety type grab-hooks, shall be furnished; and each chain shall be of an ultimate strength equal to or greater than the weight of the forklift. The driveline shall be provided with a means to disengage the wheels from the transmission for towing. Disengagement and reengagement each shall be performed in not more than 15 minutes by one man using only manual devices and common tools.

3.1.3.2 Air transportability. The forklift shall be transportable in USAF C-130, C-141, and C5A aircraft. Transportability requirements shall be in accordance with MIL-A-8421, Air Force Design Handbook DH1-11, and as specified herein. Disassembly to achieve air transportable configuration (see 3.1.2.1), shall be accomplished in not more than 1 hour and reassembly shall be accomplished in not more than 1 hour using only common tools and available lifting devices. The forklift trucks shall be capable of being driven on and off the aircraft in the transportable configuration. The truck shall be provided with a minimum of 1/2 to a maximum of 3/4 tank of fuel. All removed parts shall be transportable with the truck in the aircraft.

3.1.3.3 Rail transportability. The forklift shall be transportable by railroad in accordance with MIL-HDBK-157 and shall withstand railroad humping at speeds not less than 8 mph in both directions without permanent deformation or damage to any of its components when tested in accordance with Association of American Railroads (AAR) Rules Governing the Loading of Department of Defense material on open top cars.

3.1.3.4 Marine transportability. The forklift shall be transportable by breakbulk cargo ships, roll-on/roll-off (RORO), ships, C-8 and larger aboardship (LASH), barge carrying ships (SEABEE), LARCXV and larger amphibious vessels, and army barges and barges in accordance with MIL-HDBK-157.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of trucks from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

3.2.1 Winterized first article. When winterized forklifts are specified (see 6.2), in addition to nonwinterized forklifts, an additional forklift shall be furnished for examination and test. The additional forklift shall be a winterized forklift. Test shall be those specified in section 4 and shall be subject to surveillance and approval by the Government.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

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3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.3.1.1 Material deterioration and control. The forklift shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may affect the normal performance of the vehicle or during storage of the vehicle when stored in accordance with proper storage procedures for the type of storage involved.

3.3.1.2 Drain holes. Drain holes shall be provided where applicable, to prevent collection or entrapment of water or other unwanted fluids.

3.3.1.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.3.3 Tire material. Tires shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the forklift by the Government.

3.4 Environmental and use requirements.

3.4.1 Operating conditions. The forklift, including ROPS and FOPS shall withstand the usage encountered in military operations without damage, permanment deformation, or failure of components. The military operations include stuffing and stripping the 8 foot wide x 8 foot high family of ANSI/ISO containers at ground level and on chassis, and loading and unloading of flatcars, flatbed trailers, and aircraft with palletized loads of supplies. Operations will be on rough terrain, including unstabilized surfaces such as beaches, snow, mud, and cross country, where high flotation is needed.

3.4.2 Operating temperatures. The engine shall start without preheating within 5 minutes and the forklift shall operate as specified herein within 15 minutes after engine start in any ambient temperature in the range of -25° F to +125° F.

3.4.3 Storage temperature. The forklift shall not be damaged by storage under the conditions indicated in table I.

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TABLE I. Storage.

Air Temperature	Time Period
+160° F	1 hour daily.
+155° F	4 hours daily.
-30° F	6 hours daily (nonwinterized).
-50° F	6 hours daily (winterized).

3.4.4 Rain. The forklift shall start within 5 minutes and operate as specified herein when tested as specified in 4.6.2.28.

a. Without malfunction of any electrical component, gage, or instrument.
b. Without leakage of water in the air intake filter.
c. Without leakage of water into the torque converter oil chamber, engine crankcase sump, transmission, transfer case, instruments, gages, or fuel tank.
d. Without leakage into the cab, when furnished.

3.4.5 Winterization. When specified (see 6.2), the forklift shall be furnished with a winterization system. The winterization system shall provide necessary enclosures for engine heating and starting, battery and operator cab heating, defrosting of windshield and windows, and operation of the forklift at any ambient temperature from +125° F to -50° F. The winterized forklift shall include a cab, heaters, and winterization components to deliver, maintain, and regulate temperature. Engine operation after starting shall not be dependent on the winterization system. Controls and indicators for winterization system components shall be installed in the cab for convenient use by a seated operator. The entire forklift shall be provided with seals and hoses with manufacturer's ratings for -50° F operation.

3.4.5.1 Winterized forklift performance. With the winterized forklift stabilized at a temperature of not higher than -50° F, the winterization system shall perform as follows:

a. Warm the engine sufficiently to start within 60 minutes using an engine coolant heater.
b. Increase the engine operation temperature to the point where the engine runs smoothly without winterization or cold start aids within 30 minutes after starting the engine.
c. Permit charging of the batteries by the engine electrical charging system.
d. Increase the temperature in the cab to not less than 10° F, but not more than 30° F, at the foot treadles within 1 hour after starting the engine.
e. Defrost not less than 75 percent of each cab window within 1 hour after starting the engine, and maintain not less than 75 percent of each window free of frost with an operator in the cab.

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3.4.5.2 Winterized engine enclosure. An enclosure shall be provided to enclose the sides and top of the engine compartment. The enclosure shall be designed to prevent weather elements, particularly blowing snow and freezing rain from affecting the operation of the engine, controls, linkages, and accessories. Weather stripping conforming to MIL-R-14328 shall be used to seal around panels and doors on the sides and top of the enclosure. Other gaps in the enclosure, such as around air intakes and exhaust, shall be closed to the extent practicable with materials that will withstand the operating temperature extremes encountered in the ambient temperature range specified in 3.4.5. A manually operated air intake shall be provided to admit cooling air when the housing panels are closed. The air intake shall be located toward the end of the enclosure opposite the radiator in a position that minimizes the possibility of intake of exhaust gases. The air intake shall be designed to be opened by one man when ice, 1/4 inch thick, has accumulated on the top and vertical surfaces, as applicable. The engine housing shall be insulated on the inside with fire-resistant mastic insulation, not less than 3/32 inch thick, to reduce heat losses. Enclosure side panels shall be removable by one man. Provision shall be made for the attachment of either one 12-inch diameter duct or two 6-inch diameter ducts in accordance with MIL-H-7365 for use in directing warm air inside the enclosure. Covers for duct openings shall be hinged or removable. Removable duct opening covers shall be secured to the forklift by chains or similar means to prevent loss. All latches and handles shall be proportioned to facilitate grasping and manipulating by personnel wearing arctic gloves or mittens. Latches shall keep housing panels closed when subjected to shock and vibration; coil spring compression latches are not acceptable. The underside of the engine housing shall be enclosed with a noninsulated, removable splash pan(s). The splash pan(s) shall be equipped with service holes to provide access for scheduled maintenance. Each service hole shall be equipped with a pivoting or hinged cover plate to prevent infiltration of blowing snow when closed.

3.4.5.3 Engine coolant heater. Engine coolant heater(s) shall be provided and shall conform to MIL-H-3177. The heater(s) shall operate on fuel from the fuel tank. Heater(s) shall be located above fording depth. Any heater not located in the engine enclosure shall be enclosed in a protective metal cover or box to prevent exposure to weather elements. The enclosure shall be removable without tools and shall not affect the operation of the heater. Heaters shall be mounted on shock absorbing mounts. The coolant pump shall pump coolant from a low point at the thermostat end of the engine (the lower radiator return line, for example) through the heater to the engine cooling system at a low point on the end of the engine farthest from the thermostat end of the engine. Coolant lines shall be as short as practicable, consistent with the minimum bend radii recommended by the hose manufacturer. The routing of coolant lines shall not interfere with engine maintenance and operation. The lines shall be supported to minimize vibration. A pressure relief type safety valve with manual release shall be incorporated on the outlet side of each heater boiler to protect against excessive steam pressure and to provide a means of purging air from the coolant heater circuit. The coolant heater shall also pump heated coolant through an engine oil sump heat exchanger to provide heat to the engine oil before starting and during operation. The heat exchanger shall be located in the oil, near the bottom of the sump.

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3.4.5.3.1 Heater fuel supply system. The fuel supply system for each heater shall include a fuel pump, a filter, and a manual means to shut off fuel flow in case of emergency. The fuel pump shall be located no higher than 12 inches above the fuel tank outlet to minimize suction. Rigid fuel lines shall be installed in such a manner that they will not be subjected to chafing from vibration or stress concentrations due to sharp radii or other causes. Rigid lines shall terminate with standard, flared, or inverted flared type fittings. Flexible fuel lines shall conform to MIL-H-13444, type III, having a minimum inside diameter of 3/8 inch. Fuel lines which must be disconnected to remove the fuel pump shall be equipped with swivel or quick disconnect fittings. The lines shall be routed within or along protective framing members, away from high temperature areas, and shall be securely fastened with clips, braces, or brackets.

3.4.5.3.2 Heater exhaust system. The heater exhaust lines shall be sized to provide a minumum restriction of flow, and shall be constructed of aluminized steel pipe or flexible tubing resistant to corrosion or deterioration due to exposure to high temperature exhaust gases. The exhaust lines shall be installed to eliminate pockets which cannot be cleaned or drained and shall be positioned to eliminate discharging within the area of fresh air intakes or fogging of windows. Provisions shall be made for quick removal of exhaust sections which interfere with maintenance operations. Lines shall be insulated or located to preclude damage to other forklift components, injury to personnel or damage to clothing.

3.4.5.4 Winterized cab. Winterized forklift trucks shall be equipped with cabs conforming to 3.27 through 3.27.3, except as specified herein. Windshield wiper motors shall be electric, and shall be provided with a transmission and clutch for heavy-duty service. Levers and door handles shall be of a shape and size to permit manipulation by an operator wearing arctic mittens. Double glazing may be used for insulation or defrosting purposes, provided one pane of each double pane is hinged or removable without tools to provide access for window cleaning.

3.4.5.4.1 Cab insulation. Cab insulation shall be open cell urethane foam or polyvinylchloride insulation not less than 0.25 inch thick. The insulation shall have a nonburning rating in accordance with ASTM D 1171. The insulation shall be covered with a protective covering that presents a smooth, easily cleaned surface. The floor mat shall be fiberglass or polyvinylchloride shell insulation not less than 0.37 inch thick with an integral antiscuff skin surface. The floor mat shall be in sections, bonded to individual floor plates to allow removal of floor plate sections without disturbing adjacent areas of the mat.

3.4.5.4.2 Weather stripping. Weather stripping shall conform to MIL-R-6130, grade C, ZZ-R-765, or MIL-R-14328. When open cell material is used, it shall be covered with impregnated woven nylon or canvas fabric. The covering shall be resistant to wear and moisture. Weather stripping shall be secured to its mounting surface by a metal retaining strip. The type of weather stripping and

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the method of securing it shall prevent the entrance of blowing snow and wind and shall reduce vibration and noise to the minimum. Weather stripping shall be installed around the outer lip of the doors and the inner doorframes of the cab. The weather stripping for sealing the bottom of the door area shall be installed so that it will not be damaged by personnel stepping on it while entering or leaving the cab.

3.4.5.5 Cab heater and defroster. A cab heater conforming to MIL-H-46792, capacity B, shall be provided. The heater shall be installed outside of the cab and the air intake shall be located so that the air is drawn upward to prevent icing and clogging of the stack by snow. The heater shall operate on fuel from the fuel tank. The heater shall be located above fording depth, mounted on shock absorbing mounts. The heater fuel supply system shall conform to 3.4.5.3.1; and the exhaust system shall conform to 3.4.5.3.2. Operation of the cab heating system shall not be dependent on operation of the engine for electrical power. Permanently installed, rigid, interior ducting and plenum chambers shall be provided in the cab for directional heat toward the operator's feet and for distribution of heat from defroster nozzles over all glazing. A 24-volt booster blower shall be installed so that at least 15 cubic feet of air per minute per square foot of glass area can be blown from defroster nozzles (see AMCA Bulletin No. 210). Adjustable, flap type dampers shall be provided to regulate heated air delivery to each defroster nozzle and for control of heat directed on the operator. Air flow from the front and rear defroster nozzles shall result in a circular flow upward over the windshield and downward over the rear window. A manually operated diverter shall be provided to direct heated air to the battery enclosure only (see 3.4.5.6), split flow between the battery enclosure and the cab, or to direct flow to the cab only.

3.4.5.6 Winterized battery enclosure. Except as specified herein, the battery enclosure shall conform to 3.24.2. The enclosure shall be double walled of noncorroding aluminized steel. The chamber between the double walls shall serve as a passage for the engine coolant heater exhaust gases. The exhaust shall exit the chamber on the opposite side of the battery box from where it enters. The chamber shall not permit exhaust gases to enter the interior of the battery enclosure when the enclosure is closed. The coolant heater exhaust gases shall be used to apply heat to the battery box only during the warm up period prior to engine start. Coolant heater exhaust gases shall bypass the battery box when the engine is running. Heated fresh air from the cab heater shall enter the interior of the enclosure underneath the batteries and shall exit near the top of the enclosure. The temperature in the battery enclosure shall be automatically controlled while the cab heater is operating, so that the average temperature of the electrolyte remains between +40° F and +100° F except during the initial warm up or when the manual diverter is closed. During the initial warm up period prior to engine start, the battery heating provisions shall raise the average temperature of the electrolyte not less than 50° F in 1 hour from an initial temperature of -50° F. A drain of not less than 1/2 inch inside diameter (I.D.) and a vent shall be provided. When the battery box is installed inside the cab, the drain and vent shall be to the outside of the cab. Provision shall

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also be made for draining moisture from inside the double walls of the box. The battery box shall provide a minimum clearance of 1 inch between the sides and the batteries.

3.4.5.7 Radiator winterfront. The radiator shall be furnished with a two-piece, removable, manually controlled metal winterfront (radiator shutter) of the sliding type. The winterfront shall not jam from blowing snow or become inoperative when covered with ice up to 1/4 inch thick. In the full open position, the winterfront shall have an open area of not less than 30 percent of the radiator core frontal area. A locking device shall lock the winterfront in at least three positions: Full open, half open, and closed. Operation of the winterfront shall be accomplished without the use of tools or prybars.

3.4.5.8 Hose and hose assemblies for winterized forklifts. Hot air duct hose shall conform to MIL-H-8796, type II, class 1, or MIL-H-62028, type II, class 1. Fuel hoses shall conform to MIL-H-13444, type III. Coolant system hoses, including those for the coolant heater, shall conform to ZZ-H-428, grade A, type and class as applicable.

3.5 Safety. All exposed parts subject to high operating temperatures or energized electrically shall be located, insulated, or guarded so as to prevent hazards to operating personnel. Forklift shall comply with ANSI B56.1 and B56.6 and SAE J98. All rotating and reciprocating parts shall be guarded when such parts are exposed to contact by the operator. Protective devices shall not impair the operating functions. Steps and platforms for entering and servicing the truck shall have antiskid surfaces. For forklifts with hinged-frame (articulated) steering, a warning notice shall be stenciled near the hinge on both sides of the forklift, stating "WARNING, NO CLEARANCE FOR A PERSON STANDING IN THIS AREA", or words to that effect (see 3.32.1). Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be avoided.

3.5.1 Roll over protective structures (ROPS) and falling objects protective structure (FOPS). The forklift shall be equipped with a complete ROPS and FOPS structure conforming to SAE J1040, J1043, and SAE J231. Installation and removal of the ROPS and FOPS shall be accomplished without cutting or welding. Permanent or removable lifting eyes/devices shall be provided to facilitate removal of the ROPS and FOPS. The ROPS and FOPS shall be free from sharp edges, particularly at any point where a sling may come in contact during lifting (see 3.31). Quick disconnect for electrical lines shall be provided to facilitate removal of ROPS and FOPS. A restraint system which will retain the operator within the ROPS shall be provided.

3.5.2 Seatbelt. An adjustable, weather resistant seatbelt shall be provided with a corrosion resistant, metal-to-metal, quick release buckle conforming to SAE J386, type I and shall be anchored to the seat in accordance with SAE J383, J384, and J385.

3.5.3 Backup alarm. A backup alarm conforming to SAE J994, type A, B, or C shall be furnished. The backup alarm signal shall be audible above, and distinguishable from, the surrounding vehicle noise level.

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3.6 Human factors. The characteristics of the forklift, with ROPS and FOPS installed, shall provide for operation and maintenance by personnel ranging from the small man clothed, through the large man arctic clothed, in accordance with SAE J833 and SAE J925 and MIL-STD-1472. The operator shall be able to see the tip of at least one fork at any lift height without leaving his seat when there is no load on the forks. With forks in the carry position, the cross members of the mast structure shall not interfere with the operator's line-of-sight between horizontal and 10 degrees below horizontal using the range of eyeheights above the operator's seat. As a minimum each front wheel and the left rear wheel of the truck shall be directly visible to the operator while the operator remains in the seat in position to control the truck's drive controls.

3.6.1 Noise limits. The noise produced by the forklift shall conform with MIL-STD-1474 requirements when tested in accordance with specification, paragraph 4.6.2.27.1. The provisions of MIL-STD-1474, paragraphs 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, paragraph 4.3 and be readable at the distance indicated on the sign.

3.6.2 Exterior noise. The forklift shall meet the requirements of table 4, paragraph 5.3 of MIL-STD-1474 when tested in accordance with specification, paragraph 4.6.2.27.3.

3.7 Maintainability. Provisions shall be made for adjustment, servicing, or replacement of parts and components. When access openings are used on the exterior of the forklift, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover whenever it is required. Dimensions of hand or arm access opening shall be in accordance with SAE J925 for arctic clothing. When tested as specified in 4.6.2.37, the forklift shall have a maintenance ratio (MR) of not more than 0.15. The MR is defined as the ratio of total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components are excluded. Scheduled maintenance shall be required not more often than every 50 hours, except for before and after operation checks. Each operation in the following maintenance operations list shall be accomplished by one man (except where otherwise specified) in not more than the time specified, using common tools and special tools, if any, furnished with the forklift:

a. Remove, replace, and adjust all engine driven belts - 3/4 hour.
b. Remove and replace alternator - 1/2 hour.
c. Remove and replace all hydraulic system filters, screens, and strainer - 1 hour.
d. Remove and replace engine coolant system hoses - 1 hour.
e. Drain engine lubricating oil, remove, and replace oil filter elements and refill crankcase - 1 hour.
f. Remove and replace fuel filter elements - 1/2 hour.
g. Disconnect battery cables, remove and replace batteries, reconnect battery cables - 1/2 hour.

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h. Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill converter and transmission - 1 hour.
i. Remove and replace starter - 1 hour.
j. Bleed and adjust brakes and refill master cylinder (if applicable) - 2 men - 1 hour.
k. Remove and replace floodlight and taillight bulbs (not including blackout lights) - 1/4 hour each bulb.
l. Remove, dissassemble, service, and install air cleaner - 1/2 hour.

3.8 Reliability. The specified mean-time-between-failures (MTBF) shall be 200 hours when tested in accordance with 4.6.2.36.

3.9 Performance. The forklift shall conform to the requirements specified herein under all conditions specified without malfunction, damage, or permanent deformation. There shall be no evidence of accelerated wear or failures when operated under service conditions specified herein.

3.9.1 Stability.

3.9.1.1 Static. The rear wheels shall not clear the ground when the rated load is tilted to maximum forward tilt from maximum rearward tilt at maximum lift height under the following conditions:

a. Forklift on a hard, level surface and stationary.
b. Forklift fueled to not more than one quarter capacity.
c. No operator on truck.
d. Engine at low idle speed.

Stability characteristics shall apply to hinged-frame steer forklifts (see 3.17), at maximum steer angle.

3.9.1.2 Dynamic. The forklift shall negotiate a 30 percent slope with and without rated load in full circle operation at maximum steer angles in both directions without any wheel leaving the ground. The forklift while descending a 30 percent grade in the forward direction with rated load in the carry position at not less than 5 mph, shall stop by rapidly applying the service brake without the rear wheels leaving the ground.

3.9.2 Vehicle floatation. The forklift shall have a flotation index (FI) of not more than 25, with and without rated load. The FI shall be calculated using the following formula:

$$FI = \text{Antilog}\left[0.211 - 0.350C + 0.00710\,P - \frac{0.0752(P - R)}{P} - \frac{0.402(A_F - A_R)}{A} + 0.0346K\right]$$

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Where: P_F = Tire pressure, front (psi).
P_R = Tire pressure, rear (psi).
P = Tire pressure, average = 1/2 (P_F + P_R).
A_F = Axle load, front (lbs).
A_R = Axle load, rear (lbs).
A = Gross Vehicle weight = A_F + A_R.
K = 55.77 + 0.000139A (for nonradial tires).

$$C = \log \left[\frac{A}{0.607P + 1.35 \frac{117R}{5W + D} - 4.93} \right]$$

R = Tire ply rating.
W = Tire width (in - as cited in tire size).
D = Rim diameter (in - as cited in tire size).

The FI shall be determined with the forklift fully fueled and serviced with lubricants. For the purpose of calculating FI, the tire inflation pressures may be reduced by not more than 30 percent of the recommended pressure published in the TRA Yearbook for the forklift's maximum tire load or as recommended by the tire manufacturer at a 5 mph rating.

3.9.3 Longitudinal gradeability. The forklift shall ascend a 45 percent grade in forward gear range at a speed of not less than 2 mph with rated load.

3.9.4 Travel speed. The forklift shall be capable of travel speeds of not less than 20 mph forward and not less than 10 mph in reverse on a level surface with and without rated load.

3.9.5 Fording. The forklift shall operate in water to a depth of 20 inches. Forklift components shall be protected from damage, malfunction, and corrosion due to immersion followed by drying for those components below the fording level, and due to splash and spray for those components above the fording level. The floor of the operator's compartment may be lower than the fording level. The operator's seat cushion shall not be lower than the fording level. When tested as specified in 4.6.2.31, for a minimum of 2.5 hours the truck shall operate during and after fording operation without malfunction or damage from corrosion. The volumetric water content of lubricants and hydraulic fluid shall not exceed 1.5 percent, and there shall be no measureable increase of water in fuel at the conclusion of fording tests.

3.9.6 Load transfer. When tested as specified in test method No. 32 in a simulated container/trailer loading operation, the forklift shall transfer rated load at a rate of not less than 1 cycle per 1.2 hours.

3.9.7 Lifting speed. Speed of lift with rated capacity load shall be not less than 60 feet per minute over the entire distance from ground level to maximum fork height.

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3.9.8 Lowering speed. Speed of lowering of unloaded forks shall be not less than 40 feet per minute over the entire distance from maximum fork height to ground level. Speed of lowering forks with rated load shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level with or without hydraulic intact. The hydraulic control system shall be damped, metered, or easily manually regulated to reduce shock and prevent the truck from overturning when rated load is lowered at maximum speed and then stopped quickly.

3.9.9 Curb clearance circle. The curb clearance circle when measured in accordance with Field Test Procedure in SAE J695 shall not be greater than 39 feet when carrying rated load in the carry position for both the forward and reverse directions.

3.9.10 Upright tilt. When unloaded, truck masts shall have 10 degrees minimum forward tilt and 20 degrees minimum rearward tilt. Positive means shall be furnished to prevent cavitation of the tilt cylinders at minimum 120° F oil temperature when tilting rated load to full forward tilt at engine low idle speed. Maximum time to tilt from full rear tilt to full forward tilt at maximum fork height shall be 30 seconds at low idle engine speed. The truck shall tilt rated load at maximum fork height from full forward tilt to full rear tilt in no more than 10 seconds at governed engine speed.

3.9.11 Collapsed mast height. The collapsed mast height of an unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed 78 inches.

3.9.12 Maximum fork height. The fork height of a truck bearing rated capacity load in the horizontal position and the uprights fully extended, measured from the ground to the load carrying surface of the fork at the rated load center, shall be not less than 100 inches.

3.9.13 Free lift height. Height of forks without increase in the specified collapsed mast height shall be not less than 48 inches with rated load.

3.9.14 Low engagement height. The lowest engagement height of the forks shall be not less than 4 inches below ground level when measured at the load carrying surfaces of the forks on an unloaded truck with uprights in a vertical position.

3.9.15 Drift. The lift assembly shall be capable of holding the rated load at maximum height not less than 2 minutes with not more than 1 inch of vertical drift and not more than 0.5 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature is not less than 120° F.

3.9.16 Oscillation. Hydraulically powered oscillation of the forks shall be furnished. The forklift shall be capable of oscillating rated load at each maximum side shifted position. The oscillation of the rated load from horizontal shall be not less than 10 degrees clockwise when the load is side

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shifted to extreme left and not less than 10 degrees counterclockwise when load is side shifted to extreme right (20 degrees total oscillation). Time to oscillate the total 20 degrees shall be not more than 7 seconds at low engine idle and not more than 3 seconds at high engine idle for both clockwise and counterclockwise rotation.

3.9.17 Container operations. The forklift with ROPS and FOPS shall be capable of entering, operating within, and exiting a 20-foot long x 8-foot wide x 8-foot high container in accordance with MH5.1.1M, when the container is on the ground and when the container is supported on a chassis a minimum of 55 inches above ground. Access to the container-on-chassis shall be accomplished by traveling on a ramp in accordance with MIL-R-52899(ME).

3.9.18 Endurance. The truck shall be capable of completing 200 hours of performance on durability course as specified in 4.6.2.34 and shall also be capable of completing 2940 cycles of load placement in not more than 210 hours when as specified in 4.6.2.35 without failure, damage, or permanent deformation, and wichout exceeding specified temperature limits for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11.1), universal joints (see 3.11), or hydraulic system (see 3.12).

3.9.19 High speed. The forklift shall be capable of 4 hours of high speed operation when tested as specified in 4.6.2.33 without malfunction, damage, or permanent deformation, and without exceeding specified temperature limits for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11), universal joints (see 3.11) or hydraulic system (see 3.12).

3.10 Diesel engine. For the purpose of selecting the engine, the forklift truck shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. Unless otherwise specified herein, the engine shall meet all operational requirements and conditions specified herein without exceeding any of the operating limits specified in table I of MIL-STD-1410, or the engine manufacturer's published recommended operating limits. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer.

3.10.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the forklift under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer.

3.10.2 Engine accessories.

3.10.2.1 Starting priming system. An auxiliary fluid priming system operable from the operator's seat and conforming to MIL-E-52649, type III, shall be furnished. The primer actuator shall be a minimum of 1 foot away from the

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starter actuator. The size of the primer shall be in accordance with the engine manufacturer's recommendations. Primer shall be non-functional when engine is running. The primer may be mechanically or electromechanically operated. When the primer is mechanically actuated, it shall be furnished with a total cable length not exceeding 6 feet, and a "tee" handle which is labeled "engine primer" in accordance with the labeling requirements of MIL-STD-1472. When the primer is electromechanically actuated, it shall be furnished with a temperature control.

3.10.2.2 Alternator. A heavy-duty nominal 24-volt alternator in accordance with SAE J180 with reverse polarity protection and double lug mounting, shall be provided. The alternator shall be not less than 45-amp capacity and shall provide charge at engine low idle speed. The alternator shall be driven by the gear train, cog belt, or single or matched multiple V-belts, conforming to SAE J636 and SAE J637. The regulator shall be integral with the alternator.

3.10.2.3 Starting system. The starting system shall be 24 volts and shall include, as a minimum, a starting motor solenoid, an auxiliary relay, a starting motor, necessary wiring and cable, and starting motor switch. A means shall be furnished to prevent energizing the starter motor whenever the engine is running. The forklift shall be provided with either an interlock in the starting system or other means to prevent energizing the starter motor except when the transmission is disengaged or a mechanism to prevent the machine from moving if the engine is started with the transmission control in other than the neutral position. The starter shall be in accordance with the engine manufacturer's recommendations. The starting motor and starter solenoid shall be splashproof.

3.10.2.4 Fuel system. The capacity of the fuel tank shall provide for not less than 10 hours operation when tested as specified in 4.6.2.35 and 4.6.2.36. If the fluid level of the tank is above the engine, the tank shall be provided with a hand operated fuel shutoff valve attached directly to the tank. The fuel tank filler cap shall be captive chained. The fuel fill port shall be labeled "Diesel Fuel" with cast or embossed letters unless a data plate in accordance with 3.32.1 is furnished near the fill pipe marked with "Diesel Fuel" in not less than 1 inch letter height. The tank shall accept fuel at the rate of not less than 17 gallons per minute and shall be located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283 by a man standing on the ground. The fuel tank shall be equipped with a drainplug at the lowest point of the tank. The drainplug shall be removable with handtools without removal of any other component and shall be located such that the tank contents can be drained into a container. The fuel tank shall be protected by position or otherwise from external damage. Fuel lines shall provide continuous operation at full throttle and maximum loads, including slope ascension. The fuel system shall include a fuel transfer pump, if required to maintain engine performance, fuel strainer, fuel filter, and a fuel injection pump, including provisions for priming.

3.10.2.5 Air cleaner. The engine shall be furnished with a heavy-duty, dry type, air cleaner in accordance with MIL-A-52363, class II or class III. The cleaner shall be of the duel element or two stage type and shall include a dust

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scavenger. The capacity of the air cleaner shall be in accordance with the engine manufacturer's recommendations. An air cleaner intake restriction indicator gage shall be provided and shall be visible to the operator from the operator's seat.

3.10.2.6 Cooling system. The cooling system shall be of the closed pressure type, incorporating a radiator with a radiator pressure cap, hoses, thermostat, fan, and circulating pump. Under all operating conditions specified herein, the temperature differential between the cooling system temperature (radiator top tank) and ambient air temperature shall be less than the difference between the boiling point of the pressurized system and 125° F. Drains shall be provided to completely empty the cooling system. Petcocks on all drains shall be accessible and contents should drain into a container. The cooling system fluid shall be a solution of ethylene glycol conforming to MIL-A-46153 and water.

3.10.2.6.1 Fan. The fan shall discharge away from the operator. The fan shall rotate on antifriction bearings. The belts shall be clear of all other components within the adjustment range. Belts shall be accessible for easy replacement. The fan belt(s) shall be in accordance with SAE J636 and J637. Fan shall be protected by a mesh screen.

3.10.2.6.2 Water pump. When utilized, the water pump(s) shall be of the centrifugal type, shall be replaceable, and shall be equipped with shaft seals that will prevent leakage. The shaft shall be corrosion resistant when exposed to water and a mixture of ethylene-glycol and water. Either a lifetime lubrication system shall be required, or accessible pressure lubrication fittings shall be provided.

3.10.2.6.3 Thermostat. When utilized, the thermostat shall be in accordance with the engine manufacturer's recommendations. The thermostat shall be removable.

3.10.2.6.4 Radiator. When utilized, the radiator assembly shall be a heavy duty type and shall consist of a core with an expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator cap shall meet the requirements of SAE J164 and be accessible without removal of the engine hood. A sign warning "CAUTION - DO NOT OPEN WHEN HOT" should be placed near the radiator cap. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced in normal operation of the forklift. The radiator shall be protected by a heavy grilled guard or by the counterweight. A shield shall be provided, if necessary, to protect the radiator from mud and gravel thrown by the tires. Hoses shall clear the forklift structure and other components under all operating conditions specified herein. The cooling systems shall include baffles and shroud to prevent recirculation of cooling air.

3.10.2.6.5 Gaskets, seals, and packing. When utilized, gaskets, seals, and packing used in the cooling system shall be made of materials capable of resisting deterioration from mixtures of water and ethylene-glycol-base anti-freeze conforming to SAE J1034 or MIL-A-46153.

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3.10.2.6.6 Engine coolant hose. When utilized, all engine coolant hoses shall be SAE J20, heavy-duty type. All hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, type F, style 1, 2, or 4.

3.10.2.7 Governor. A governor shall be provided to limit the engine speed to not more than the intermittent duty speed recommended by the engine manufacturer when the forklift is operated under any condition specified herein.

3.10.2.8 Exhaust system. The exhaust system shall be protected against entry of rain. The back pressure of the exhaust system, including muffler, shall in no case exceed the maximum recommended by the engine manufacturer when measured approximately 2 inches beyond the header outlet at any load up to rated net continuous load. The exhaust system shall be arranged or shielded to prevent burns to the operator and maintenance personnel while working on the step surfaces of the forkfit. Exhaust discharges shall be directed to minimize fumes at the operator's station or obstruct the view of the operator. The exhaust system shall be provided with a spark arrestor conforming to Forest Service Standard 5100-1 and SAE J530 for not less than 80 percent of efficiency. The spark arrestor shall be provided with a means for cleaning out trapped particle.

3.10.2.9 Crankcase ventilation system. The crankcase ventilation system shall prevent water entering the engine during fording operations. When the crankcase vapors are directed into the induction system, means shall be provided on turbocharged engines to prevent coking of turbocharger blades. No vapors or condensation shall be deposited on or come in contact with the air-cleaner element.

3.10.2.10 Oil filter. Oil filter(s) of the heavy-duty full flow type or heavy-duty full flow and partial flow type shall be furnished in accordance with the engine manufacturer's recommendations. A label specifying filter type shall be placed near the point of filter installation. Oil lines, when applicable, to and from the filters shall be installed to minimize vibration and shall be of flexible hose meeting the requirements of SAE J30(d) with reusable fittings.

3.10.2.11 Tachometer drive. A heavy-duty key type tachometer drive shall be provided and shall have a 5/32-inch female connection in accordance with SAE J678 for diagnostic purposes. The speed of the tachometer drive shall be 1/2 of the crankshaft speed.

3.10.2.12 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained metal threaded cap which provides a positive seal of the sampling port. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. The sampling valve shall either be labeled or a decal shall be provided adjacent to the valve indicating the type of oil sampled (engine or transmission, whichever is appropriate).

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3.11 Drive assembly. The drivetrain assembly shall consist of all components necessary to transmit power from the engine to all four wheels. All rotating shafts and axles shall be supported on antifriction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system shall be provided for other friction parts. Gear housing shall be equipped with filler plugs and magnetic drainplugs. Differential ring and pinion gears shall be adjustable. Drive shafts shall be provided with a minimum of two heavy-duty universal joints and a nonbinding slip joint. Drivetrain components shall conform to and shall be installed in accordance with each component manufacturer's recommendations. Drivetrain universal joints shall not exceed 180° F when the forklift is tested as specified herein.

3.11.1 Transmission. The forklift shall be provided with either a hydrodynamic or hydrostatic transmission as defined in SAE J645. The forklift shall be capable of accelerating from a standstill to maximum speed under continuous full engine throttle on level ground and shall be accomplished without damaging the transmission. The transmission shall be protected from damage when the operator moves or attempts to move the directional control to the opposite direction while the forklift is in motion. The transmission oil shall conform to MIL-L-2104, grade 10, or MIL-L-46167. The fill tube shall be functionally labeled "transmission fill". All transmission component material shall be compatible with these oils. The transmission fluid shall be filtered by filter(s) having replaceable elements in accordance with the transmission manufacturer's recommendations. The transmission shall be provided with a disconnect control. The disconnect control shall disengage the transmission and apply the service brakes while allowing the engine to be operated throughout the entire speed range. The forklift shall be capable of achieving specified lifting, lowering, and tilting rates of the rated load while the disconnect control is engaged and transmission control is in forward position, without moving the truck.

3.11.1.1 Hydrodynamic transmission. The transmission shall provide controlled application of the clutches to provide smooth shifting, limiting slippage which would damage the clutches. All transmission gears shall be constant mesh. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a mechanically driven pump. The transmission shall be capable of withstanding maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 250° F with an initial fluid temperature of not less than 157° F in an ambient temperature of not lower than 25° F.

3.11.1.2 Hydrostatic transmission. The hydrostatic transmission shall be a closed loop type with a reversible variable displacement pump and a reversible variable or reversible fixed displacement motor. A charge pump shall be provided and shall be gear or shaft driven by the engine and shall provide sufficient oil under pressure to retard cavitation of the pump and motor and to maintain a constant, smooth control lever response independent of operating conditions performed at full throttle engine speeds. With an oil temperature of 140° F $\pm$5° F and new filter, the vacuum at the charge pump inlet port shall not

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exceed 5 inches Hg when tested in accordance with 4.6.2.21.1. A shuttle valve or equivalent means shall be provided to automatically prevent loss of high pressure oil from the closed loop pump/motor circuit during the transition of rapid change in direction of the forklift. A set of reduction gears with a minimum of two speed ranges capable of being shifted on the move under full engine torque may be used in conjunction with the hydrostatic transmission. When a gear range selector is provided it shall be accessible to the operator in the normal operating position and shall be labeled to indicate speed ranges. Oil temperatures in the transmission shall not exceed 180° F when measured at the pump case drain and motor case drain under any operating conditions specified herein. When a heat exchanger is required to meet temperature limits stated herein, means shall be provided to insure recommended case pressures of the pump and motor are not exceeded during all operating conditions specified herein including cold weather start-up. The transmission shall be capable of withstanding maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 180° F. The system shall withstand all operations specified herein without permanent deformation, damage, or leakage.

3.11.2 Rear axle oscillation. The rear axle shall oscillate freely such that the total vertical travel of each rear wheel shall be a minimum of 12 inches (six inches above and six below the horizontal axle centerline).

3.11.3 Traction control. The front differential shall be of the automatic locking or driver controlled locking type to insure power is transmitted to the wheel(s) having traction when the opposite wheel(s) loses traction or is off the ground. If a driver controlled locking differential is used, an engaged warning light labeled "front differential is locked" (in accordance with MIL-STD-1472) shall be installed in a position clearly visible to the driver to prevent axle damage if the vehicle is driven on high coefficient surfaces and shall automatically disengage after release.

3.12 Hydraulic system. The hydraulic system shall consist of all fluidpower components necessary for the operations of the forklift, including the steering system and hydrostatic transmission system but does not include the brakes or the hydrodynamic type power train. All filter tubes shall be labeled. The working hydraulic system (lift, tilt, fork functions and steering) maximum operating pressure (relief valve setting) shall not exceed 3000 psi and if furnished the hydrostatic transmission relief valve setting shall not exceed 6100 psi. The system fluid shall conform to MIL-L-2104, grade 10 or MIL-L-46167. All hydraulic system component material shall be compatible with these oils. The temperature of the hydraulic fluid shall not exceed 100° F above the ambient during all operations and test specified herein. Hydraulic fittings on pumps, valves and cylinders shall be either the O-ring, straight thread type, as specified in SAE J514 or the 4-bolt flange type as specified in SAE J518 for 1 inch and smaller, and only the 4-bolt flange type as specified in SAE J518 for ports larger than 1 inch. The hydraulic system shall be flushed free of foreign matter including weld slag and spatter. The connection ports of the hydraulic components shall be sealed by the component manufacturer. The hydraulic system

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shall permit the rated load to be lowered at the rate of speed specified in 3.9.8 in the event of failure of or damage to the hydraulic line supplying the cylinder. A straight thread O-ring boss diagnostic port conforming to SAE J514 shall be provided at the pump(s) outlet or pressure inlet of each control valve.

3.12.1 Hydraulic lines. Hydraulic lines shall consist of tubing, flexible hose and steel fittings. Hose or a combination of hose and tubing shall be used between components that are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance, maximum protection and provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fitting, and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines, other than suction lines, and cylinder bleeder lines, shall be considered pressure lines. All lines shall be flushed free of foreign matter.

3.12.1.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 or MIL-H-52471. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. The truck shall be compatible with hose assemblies with field attachable reusable fittings in place of each hose assembly with permanently attached couplings when tested as specified herein. All fittings shall be 37-degree flare, female swivel or 4-bolt split flange as specified in SAE J516. Hose ID shall be limited to 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. The hose selected shall have a rated operating pressure that is greater than the hydraulic circuit maximum relief valve setting. The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.12.1.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall meet the requirements specified herein. Pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. Tubing shall have a minimum burst pressure that is equal to or greater than 3.0 times the system maximum working pressure. Tubing shall be securely clamped. All bends shall be smooth without flattening, kinking or wrinkling of the tube. The term "without flattening" is defined as follows:

> Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the nominal tube diameter.

3.12.1.3 Pressure tube fittings. Unless otherwise specified herein pressure tube fittings shall be 37-degree flared and straight thread fittings conforming to SAE J514 or 4-bolt split-flange type conforming to SAE J518. Jump size tees may be used. The 4-bolt split-flange type connections may terminate in either a flange head that incorporates a SAE J518 plain mating face or a flange head that integrally incorporates four boltholes and an O-ring groove for mating to a SAE J518 port face. Pressure tube fittings that incorporate the SAE J518 connection

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shall contain a boss into which the tube shall be brazed or welded. Pressure tube fitting sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inch. Pressure tube fittings shall have a minimum burst pressure 3 times the maximum operating pressure.

3.12.1.4 Suction lines. A pump inlet suction line shall include provision for flexibility when there is vibration or movement between the reservoir and the pump. The suction line shall be sized so that the pump inlet pressure does not drop below atmosphere by more than 5 inches of mercury vacuum at 140° F +10° F oil temperature and at governed engine speed. The inlet suction line installation shall include no stress concentration at the end fittings and no kinks. Hose shall conform to SAE J517, SAE 100R4 or shall be oil resistant hose. Suction hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, type F, style 4 or saddle worm thread type with stainless steel inner liner, a heavy duty T-bolt clamp or shall be in accordance with fittings specified in 3.12.1.1.

3.12.1.5 Quick disconnect couplings. Quick disconnect hydraulic couplers shall be used in hydraulic line where separation is made for transportability requirements and they shall perform as specified herein. The hydraulic couplings shall connect to lines equal to or smaller than 1 inch with straight thread fittings conforming to SAE J514 and to lines larger than 1 inch with the SAE J518, 4-bolt, split flange face fittings. The coupler halves shall be restrained to prevent contact with the ground or any surface of the boom when connected and when disconnected. The coupler halves shall be identified with metal identification plates, metal tags, metal bands or by color coded anodized aluminum fittings to assure assembly of correct mating halves. Each coupler half shall be provided with a captive chained cap to protect the opening when the coupler is disconnected. The caps of the mating coupler halves shall fit together when not in use to prevent corrosion, damage, and contamination of the mating surfaces. The hydraulic coupler shall not have any partially coupled unlocked positions in which the couplings can remain stable and permit flow. After 50 connections and disconnections, the coupler shall meet all requirements specified herein. When coupled and uncoupled with 60 psi or relief valve pressure, whichever is greater, is applied, the coupler shall seal the ends of the disconnected lines at the point of disconnection and shall not permit external fluid loss exceeding 1/2 fluid ounce. At full line flow, the pressure drop across the quick disconnect coupler shall not exceed 15 psi at 120° F. The connected couplings and disconnected halves shall withstand a surge pressure equal to 150 percent of the relief valve setting without leakage or rupture. The quick disconnect couplings shall show no leakage and shall be in accordance with ANSI B93.42 and meet all requirements specified therein.

3.12.1.6 Return lines. Return lines shall be in accordance with SAE J517.

3.12.2 Hydraulic pumps. The hydraulic pump(s) shall have been tested in accordance NFPA Recommended Standard T.3.9.18-1976. When tested in this manner, the contaminant tolerance profile for the hydraulic pump(s) shall either:

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a. Not fall below or cross the B_{10} curve of the hydraulic system filter when tested in accordance with MIL-F-52723 (see 4.6.2.21.2).
or else
b. The pump Omega rating shall be less than the filter Beta 10 rating.

3.12.3 System filtration. System filter(s) shall be furnished either in the pressure line, return line (suction line of charge pump allowed for filtration of hydrostatics), or combinations (as specified in 3.12.3.1 or 3.12.3.2).

3.12.3.1 Return line filters. Although the system filter element need not be qualified to MIL-F-52723, it shall meet the performance requirements as stated therein. The filter shall maintain a full flow condition for the duration of reliability testing as specified in 4.6.2.36. The filter bypass may open for system oil at a temperature less than 70° F. The filter element shall have been tested to the performance requirements of MIL-F-52723 at a flow rate equal to or greater than the truck maximum system oil flow at maximum governed engine speed. All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed without the removal of other components for examination or replacement without spilling or draining oil from the reservoir. At the maximum system oil flow through the filter, including lowering rated load at maximum governed engine speed and an oil temperature of 150° F ±5° F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the system filter and shall open at not less than 10 psi greater than the pressure drop measured across the new filter assembly and it shall permit full flow at a pressure drop not greater than 35 psi. The filter housing shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage. The system filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the filter element is clean and when the system filter is beginning to bypass, requiring a change of the element.

3.12.3.2 Pressure filter. The pressure filter(s) shall conform to 3.12.3.1 except that proof pressure test shall be 150 percent of the maximum system operating pressure.

3.12.4 Reservoir. A hydraulic reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The reservoir shall conform to SAE J614b and shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The Add and Full marks on the dipstick or sightglass(es) shall indicate the correct oil level when the truck and forks are on level ground, and the oil is at normal operating temperature. The fillercap and tube shall be

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located to provide for filling from a standard 5 gallon container conforming to MIL-C-1283 by a person standing on the ground or a surface on the forklift intended for standing, and to prevent entrance of contaminants thoughout filling and checking operations. The reservoir filler neck shall contain a removable 50 mesh, or finer strainer retained in place unless all oil added to the reservoir is filtered by an equal or finer filter prior to passing through the pump. The filler tube shall be not less than 1 inch inside diameter. The reservoir pump inlet line suction openings except for the auxiliary steering pump opening, shall be located a minimum of 2 inches above the bottom of the reservoir. When furnished, the auxiliary steering pump inlet line suction opening shall be located a minimum of 1/2 inch above the bottom of the reservoir. Except for sealed reservoir, a pressurized breather with a 30 psi maximum relief valve and 1 inch Hg vacuum breaker shall be furnished. The breather shall have a nominal rating of 40 microns. The breather shall preclude the entrance of water. The reservoir shall provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow except the vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir. When filters are located within the reservoir, a removable 50 mesh, in-tank suction strainer with bypass valve shall be installed in the suction line of the reservoir. The reservoir shall have suction flanges and one or more access holes with covers for inspection, cleaning and maintenance of the entire inside.

3.12.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be generated only when a cylinder is bottomed or when the system is subjected to shock loading.

3.12.6 Hydraulic control valves. The control valve spools shall be spring centered with sufficient force to return and retain the levers at the center position throughout all forklift operations. The spool connection eye end shall conform to SAE J748 and shall not bind while operating as specified herein. The exposed control-valve spool surfaces shall show no evidence of corrosion or pitting when tested as specified herein. The contractor shall verify that the control valves were tested in accordance with procedures specified in SAE J747 and J1117. The inlet to outlet port pressure differential shall not exceed 3 percent of the maximum operating pressure at rated flow.

3.12.7 Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seamless steel tubing or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. All hydraulic system and steering cylinders shall perform as specified herein without buckling or bending. Each piston shall have a bearing surface concentric with the piston to carry the piston side loads against the

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cylinder walls. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder. All cylinder rods shall be hard chrome plated. All cylinders shall withstand a pressure that is equal to 1/5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, internal or external leakage.

3.12.8 Contamination levels for hydraulic system. When tested as specified in 4.6.2.21.3 particle counts shall not exceed 1000 particles per milliliter greater than 10 micrometers and particle counts shall not exceed 20 particles per milliliter greater than 20 micrometers.

3.12.9 External leakage. After completing tests specified in 4.6.2, the degree of external hydraulic system leakage shall not exceed a class 4, for dust-free conditions and a class 4D for dusty conditions of SAE J1176.

3.12.10 Hose reels. When hydraulic hose reels are used on the forklift to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges.

3.13 Mast and fork carrier assembly. Mast uprights shall be furnished and shall be of the telescopic, roller type. The lift shall be hydraulically actuated, and the lift cylinder shall be mounted and positioned so that it does not protrude forward of the plan outline of the uprights. Chains shall be furnished and shall be installed in such a manner that they cannot be displaced from the roller until they have been raised a minimum of one-half the chain side plate height. Chains used in the lifting mechanism shall have a factor of safety of not less than 4.6 to 1 based on average ultimate strength. Adequate means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Carriers shall be equipped with adjustable thrust bearings or side rollers to provide for forces imposed by off-center loading. The forks and fork carrier shall be positioned on the lift mechanism in such a manner, that a load wider than the forklift, positioned at the rear of the forks, will not come in contact with the front tires or any other portion of the forklift in any possible fork position. The fork carrier shall not come in contact with the front tires or any other portion of the forklift in any possible fork position. The fork carrier shall not exceed 40 inches in height.

3.13.1 Sideshift. The lift mechanism shall be equipped with a hydraulically powered sideshift mechanism that shall provide lateral displacement of the rated load from a centered position to either side, not less than (0.5W-18) inches, where "W" equals the forklift width. Fork carrier width shall not exceed the width of the forklift. The sideshift function shall provide for full sideshift travel specified, with the forks positioned at any point within the extreme of the fork spread specified. When the forks are sideshifted to either maximum sideshift position, the truck width shall not be increased by more than 1 inch. There shall be no protruding bolts or appendages beyond the side plane of the sideshift carriage assembly. Time to sideshift rated load horizontally

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from one extreme to the other extreme shall not exceed 15 seconds at engine low idle and shall not exceed 5 seconds at engine high idle. The sideshift mechanism shall have sufficient capacity to sideshift rated load uphill at maximum oscillation position of the carriage. No permanent deformation or damage shall result when the sideshift mechanism is operated a minimum of 100 cycles when tested in accordance with 4.6.2.23. At full sideshift to either side, the forklift should be required to withstand without permanent deformation or damage 300 percent of the rated load on the forks at a 24-inch load center.

3.13.2 Load backrest. A metal back rest, removable without disturbing any other component, shall be provided. Load backrest shall be flush with the forward vertical surface of the forks. In conjunction with fork and hanger design, it shall provide a vertical rear guard at least 20 inches high measured from the load carrying surface of the forks and at least equal to the width of the carriage. It shall have strength and stiffness sufficient to prevent a rated load 48 inches high or less from forcing backrest against the uprights when the mast is at maximum rearward tilt. The backrest shall be firmly attached to the lift carriage. It shall not unduly obstruct operator's visibility. The spacing of vertical members shall be no more than 6 inches between members. The backrest shall be capable of withstanding, without permanent deformation, a horizontal static test load equal to the rated load uniformly distributed over the area from the load carrying surface of the forks to the top of the backrest. The removable backrest may be eliminated if fork and hanger assembly will provide protection equal to the above.

3.14 Forks. The forklift shall withstand, without permanent deformation or damage, 300 percent of the rated load on the forks at a 24-inch load center, with the fork carrier centered; and 200 percent of the rated load on the forks at a 24-inch load center at full shift to either side. Forks shall be attached to the fork carrier in a manner which prevents lateral movement of the forks, except when fork position is intentionally changed by the operator. Fork positioning shall be manual. The fork carrier shall permit the forks to be moved together to an inside-to-inside dimension of not more than 4-inches (between the inside edges of the forks) and moved apart as far as an outside-to-outside dimension of not less than 34 inches (over the outside edges of the forks). The fork length shall be 40 inches +0, -1/2 inch. The fork width shall be not more than 6 inches. The fork thickness shall be not more than 1.5 inches at the heel tapered to not more than 0.5 inch at the tip. The taper shall be on the underside of the forks and shall be not less than 14 inches long. The fork tips shall be chamfered or rounded to break corners and provide smooth edges.

3.15 Wheels and tires.

3.15.1 Wheels. The forklift shall be equipped with four demountable wheels. The wheels shall be fastened to the hubs in accordance with the axle manufacturer's recommendations for off-road applications. Wheel rims shall be in accordance with the TRA Yearbook.

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3.15.2 Tires. All tires shall be new and unused, wide base, off-the-road type, and of identical size and construction. The tires shall be pneumatic, low pressure, tubeless, furnished with type L-2 traction-type tread or radial tires equivalent to the type L-2 traction-type tread. Tire loadings shall be in accordance with table WD-1C tables of the TRA Yearbook, or rates approved by the rim or wheel manufacturer and the tire manufacturer at the 5 mph service rating. Valves and caps shall be in accordance with the TRA Yearbook. Large bore valves shall be furnished. A valve adapter shall be furnished to reduce the large bore to the standard passenger vehicle size bore. Valve caps shall be metal with elastomer innerseal. Hydroinflated tires are not acceptable.

3.16 Wheel guards. Wheel guards shall be provided for all four wheels to prevent foreign objects and road splash being discharged in the direction of the operator. Each guard shall support a weight of 300 pounds without permanent deformation or damage.

3.17 Steering system. The steering method shall be either the hinged-frame (articulated) type or Ackermann type providing the operator with two-wheel rear steering or four-wheel crab steering. The steering shall be actuated by hydraulic cylinders. The steering system shall permit control of the forklift while traveling at any speed up to the maximum specified travel speed (see 3.9.4). Ackermann type steering shall be provided with an adjustable, mechanical followup mechanism to maintain a positive ratio between the amount the steering wheel is turned and the amount the forklift turns, except that a followup need not be provided for a hydrostatic steering system. The power steering actuating device shall be located so as to absorb all road shock without transmitting the shock effect to the steering wheel. The forklift shall operate without loss of operator control when tested as specified in 4.6.2.12.

3.17.1 Steering control. The steering system shall be controlled by a steering wheel. The effort required to turn the steering wheel shall not exceed 14 pounds pull at the rim. The steering wheel shall require not more than 4-1/2 turns to move the wheel from one stop to the opposite stop. The steering wheel shall require not more than 5 seconds for the maximum range of movement at low idle engine speed and shall require not less than 1.5 seconds at governed engine speed. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). The steering wheel shall be capable of withstanding a minimum 300 pound force applied perpendicular to the plane of the steering wheel, and a minimum 150 pound force applied tangentially to the rim of the steering wheel, without permanent deformation or damage. In case of power steering assist failure, the steering gear shall afford the operator sufficient mechanical advantage to guide the truck during an emergency stop (not more than 50 pounds turning force shall be required in the event of power assist failure).

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3.17.2 Emergency steering. Emergency steering conforming to SAE J53 shall be provided for retaining steering control in the event of engine failure or steering power source failure when traveling at any speed to the maximum speed.

3.18 Brake system. Service and parking brakes shall be provided and shall be independently actuated systems. If a separate hydraulic system for the brakes is utilized, the brake system shall be filled with a fluid that is listed on the qualified product list of MIL-B-46176 unless the fluid used is petroleum base non-hygroscopic oil conforming to MIL-L-2104. If air brakes are provided, air brake service and emergency line coupling conforming to SAE J318 shall be provided on the rear of the vehicle.

3.18.1 Service brakes. Foot controlled service brakes shall be provided for all wheels. The brakes shall be capable of bringing the forklift, with rated load on the forks, to a complete stop without any wheel leaving the ground from a speed of 15 mph as follows:

a. Dry - within 25 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.
b. Fade - within 25 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.
c. Wet - within 35 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.

Pedal force while making any stop shall not exceed 90 pounds. The brake system shall provide for bleeding of air from hydraulic lines and for draining of moisture from the compressed air system, as applicable. Automatic or manual brake adjustment provisions shall be provided. Automatic adjustment shall maintain correct lining to drum or disk adjustment without looseness or lockup. Automatic adjuster shall be enclosed or guarded for protection from environmental damage or malfunction. The brake system shall withstand an operator's brake application force of not less than 250 pounds at the pedal, a minimum of 10 times, without permanent deformation or damage.

3.18.2 Parking brake(s). Parking brake(s) shall hold, but not necessarily stop the forklift without rated load on the forks when the forklift is stopped on a 30 percent grade in both the ascending and descending position. The parking brakes shall be hand operated and equipped with a locking device. The manual control shall be located within reach of the operator and in a noninterfering position. Hand lever pull to set the parking brake shall not exceed 90 pounds. The parking brake shall be an independent mechanical friction brake mounted on the drive or propeller shaft rotating with the wheels, or a separate isolated system actuating the service brakes. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any assembly.

3.18.3 Emergency braking provisions. The forklift shall provide not less than six "dead engine" stops with rated load while descending the 30 percent grade. The emergency braking provisions may be provided through either the parking brake or service brake system.

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3.19 Operator access. Operator access systems to the forklift shall be in accordance with minimum requirements of SAE J185. Access and egress to the operator's compartment shall not require the operator to climb any higher than 6 inches above the floor of the operator's compartment.

3.20 Operator's seat. A cushion seat and backrest shall be furnished in accordance with the minimum requirements specified in SAE J899 and located in accordance with SAE J1163 to permit 90 percent of U.S. male/female operator population (SAE J833) to reach controls in the optimum areas of the forklift as defined in SAE J898 by operator clothed in NBC and cold-wet gear. The seat shall be easily adjustable horizontally and vertically (not to exceed 3 inches) by the operator without use of hand tools so the operator can see the tip ends of the forks when attempting to pick up pallets. The seat cushion and backrest shall be foam rubber or polyurethane, covered with slip covered vinyl upholstery having a minimum weight of 40 ounces per yard for 54-inch width.

3.21 Hood. The engine shall be protected by a hook cover and sidepanels, except that sidepanels may be omitted when wheel guards provide the engine compartment with protection from road splash and debris. Either the hood cover or sidepanels shall be quick opening if both are provided. The hood cover, if lift up type, shall be held in the open position in a safe manner by counter-weight or spring-and-linkage mechanism or latch or by a brace which cannot be inadvertently released while servicing the forklift. The hood cover or side-panels shall be capable of being opened or removed by one man and shall be within the design weight limits of MIL-STD-1472, weight requirements. Handtools shall not be required for release of quick opening hood or sidepanels. Elastomer hood latch or paddle latch shall be provided on the hood cover and sidepanels. All engine accessories shall be completely accessible, without requiring the removal of any additional forklift structure, when hood cover and sidepanels are open.

3.22 Instrument panel. The instrument panel shall be located so that the operator can read the instruments from his normal operation position. The instrument panel shall be installed on an incline at a minimum of 5 degrees to a maximum of 80 degrees. On this panel shall be mounted the charging system monitor, engine oil-pressure-monitor, engine-coolant temperature monitor, torque converter for transmission-fluid temperature monitor, fuel gage, and any other instruments used, unless otherwise specified herein. Universal symbols shall be furnished in accordance with SAE J298 and ANSI MH11.3. All gages shall be flush mounted and shall be provided with switch-controlled illumination. A lateral level indicator for the truck shall be provided on the instrument panel.

3.23 Controls and instrumentation. Each control shall be provided with symbolic labels either on plates conforming to 3.23 or embossed or depressed on or near the control. Labeling for levers which control any part of the lift mechanism shall conform to ANSI MH11.3. The labels shall be visible to the operator from his normal operating position. All controls for the forklift shall be located in the operator's compartment and shall be within the maximum area specified in SAE J898. Unless otherwise specified herein, the controls

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shall conform to MIL-STD-1472. Electric controls and instrumentation shall be weather-tight. All hand operated controls shall be operable by persons wearing cold-wet gloves. Mechanical push-pull remote controls, when furnished, shall be of waterproof, heat resistance, antifriction type, conforming to requirements for marine cables specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cables shall be fabricated of corrosion-resistant steel that has been passivated as per QQ-P-35. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. The "as-installed" mechanical efficiency of each installation shall be not less than 70 percent and the no-load friction shall not exceed 5 ounces per foot. The control manufacturer's recommended load ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Only controls associated with the cab, such as windshield wiper and heater controls, may be mounted on the cab structure.

3.23.1 Light switches. "On-off" light switches shall be furnished on the instrument panel. One switch shall control the forward floodlamps and one switch shall control the rear floodlamps as specified in 3.26.2. One light switch conforming to MS51113 and clearly labeled as to its specific function (i.e., front headlights, rear flood lamps, etc.) shall be provided for the floodlamps (see 3.26.2), rear taillights (see 3.26.2.2), blackout headlight (see 3.26.2.1), two blackout stoplight-taillights (see 3.26.2.1), and instrument panel lights (see 3.22).

3.23.2 Engine run control. An engine run control shall be furnished and shall be lever type and waterproof. It shall have not less than two positions: "on and off". The switch shall control operation of monitors, and electrical accessories, except horn.

3.23.3 Starting system control. The starting system solenoids (see 3.10.2.3), shall be operated by a pushbutton starter switch. The starter shall be inoperable when the engine run control is in the "off" position.

3.23.4 Braking and inching pedal(s). Two pedals shall be provided for declutch or inching and braking control. The right pedal located for convenient right foot operation shall provide braking control only. The left pedal located for convenient left foot operation shall be a combination declutch or inching and braking pedal. The effort required to activiate the inching or declutch control shall be not less than 25 pounds. Each pedal shall be a minimum of 3 inches wide and shall have an all around clearance of not less than 1-1/2 inches. Grease fittings or permanent lubrication shall be provided for lubricating moving parts of all pedals and shafts.

3.23.5 Accelerator control. The accelerator control shall be installed for comfortable right foot operation and shall be located to the right of, and shall be free of interference from the brake pedal. The accelerator shall have an all around clearance of not less than 3 inches.

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3.23.6 Transmission controls. Selective forward and reverse and speed range controls shall be provided and shall be located for convenient left hand operation. Forward, reverse, and speed ranges shall be labeled in accordance with SAE J298. Movement of the control lever in a forward or upward direction shall cause forward motion of the forklift. Transmission control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operation specified herein.

3.23.7 Hydraulic controls. Lift, tilt, oscillation, and sideshift, hydraulic controls shall be self centering lever(s), located for convenient right hand operation. All control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operations specified herein. The force necessary to move any control to the end of its travel shall not exceed 20 pounds and shall have a minimum force of 2 pounds at the handgrip. Total control lever travel from one end to the other shall not exceed 14 inches, measured at the center of the handgrip. Clearance between control handgrips shall be not less than 2 inches in any control position. When separate lift and tilt levers are used, the tilt control shall be at the right of or below the lift control, as viewed by the operator. Control functions and predominant control motion shall be as follows (function listed, followed by control motion - the first applicable motion listed, depending on lever mounting, shall apply):

a. Lift load - toward the operator, rearward, or upward (opposite direction to lower).
b. Tilt load rearward - toward the operator, rearward, or in the same rotational direction as fork movement (opposite direction to tilt forward).
c. Sideshift to right - to the operator's right, rearward, or upward (opposite direction to sideshift left).
d. Oscillate clockwise (as viewed by operator) - to the operator's right, rearward, or upward (opposite direction to oscillate counter-clockwise).

3.23.8 Electrical system monitor. A remote indicating ammeter, or voltmeter, or alternator indicator red light shall be furnished. When furnished the ammeter shall be of sufficient capacity to indicate variations in current supply or demand experienced by the forklift, exclusive of starter requirements. When furnished, the indicator light shall light when the alternator is not charging and the engine run control is "on" and shall be labeled "alternator" or with appropriate universal symbology.

3.23.9 Engine oil pressure monitors. A remote indicating engine oil pressure gage and a low pressure red indicator light shall be furnished. Means shall be provided to check operation of the light.

3.23.10 Engine coolant temperature monitor. A remote indicating engine coolant temperature gage or a high temperature red indicator light shall be furnished. If an indicator light is furnished, means shall be provided to check operation of the light.

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3.23.11 Transmission temperature monitor. A remote indicating torque converter or transmission temperature gage or a high temperature red indicator light shall be furnished. If an indicator light is furnished, means shall be provided to check operation of the light.

3.23.12 Hourmeter. An operational hourmeter which registers the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hourmeter shall be accessible for reading without moving any component of the truck.

3.23.13 Fuel gage. An electric, thermally stabilized, remote indicating fuel gage shall be provided.

3.23.14 Air pressure monitor. When a compressed air system is furnished, an air pressure gage shall be furnished and an audible alarm shall indicate when air pressure is inadequate for proper brake operations.

3.24 Battery and battery mounting.

3.24.1 Batteries. A minimum of two 12 volt batteries shall be furnished. Batteries shall be in accordance with MS35000, type 6TN and shall be readily accessible for service. The batteries shall be shipped charged and dry. Electrolyte shall be furnished in a separate container as specified in MIL-STD-162. The batteries shall be negative grounded in accordance with SAE J538. The battery terminals shall be provided with guards to prevent accidential shorting.

3.24.2 Battery mounting. Battery mounting shall be in a weathertight compartment having a no-sag cover. The compartment shall be accessible for battery removal without chafing or other damage to the battery. Batteries shall be located so they can be cleaned, serviced, and removed without removing any component except the quick-release battery box cover if a battery box is furnished. Battery supports, holddown, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with a coating conforming to TT-C-494, type II. The battery shall be mounted in such a manner as not to interfere with access to engine components (accessories). Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall not be designed for or dependent upon special battery shapes and shall be provided to hold the battery in a fixed position. The battery compartment shall have provision for drainage, and shall have provision for gas venting at or near the top of the compartment. No-sag cover and positioning shall be protected against short circuiting. The cables shall be protected by rubber grommets or insulated passages at entry to the battery box.

3.24.3 Battery cable terminals. Battery cables shall be in accordance with SAE J541. Positive and negative cable terminals shall be identified. Terminals secured with a plated corrosion resistant bolt and nut shall be provided on the

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ends of cables connected to the battery to facilitate disconnection and battery removal.

3.24.4 Slaving components. The truck shall be equipped with a 24-volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the truck batteries and slave starting of the engine from an external power source and shall also provide a power source for charging and slaving other equipment. The slave receptable shall be installed on the exterior of the truck near the battery enclosure and shall be accessible to personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate furnished adjacent to the receptacle which reads "24-volt slave receptacle". The plate shall be in accordance with 3.32.1.

3.25 Pintle hook. A rear mounted pintle hook conforming to MS51335 shall be provided. The pintle hook shall be located on the vertical centerline of the forklift not less than 31 inches nor more than 37 inches above ground level. The pintle hook mounting shall be of sufficient strength to withstand, without permanent deformation or damage, the drawbar pull induced when the pintle hook is restraining forward motion at a torque converter stall and when used in its normal operations.

3.26 Electrical components and circuits.

3.26.1 Circuit breaker. Each electrical circuit shall be protected with a circuit breaker in accordance with SAE J553. Fuses are not acceptable.

3.26.2 Lights. The forklift shall be provided with not less than four floodlamps for forward illumination and not less than two floodlamps for rearward illumination. The lights shall be shock mounted in elastomer ring housings. The floodlamps shall be Trade No. 4578 in accordance with SAE J598. All forward floodlamps shall be positioned such that they are visible by a direct line of sight from an observer positioned not less than 20 feet forward of the truck when rated capacity load (see 6.9.2) is in carry position (see 6.9.3). Two forward floodlamps shall be mounted on the fenders and two forward floodlamps shall be mounted under the ROPS near the front of the truck. The floodlamps mounted on the ROPS shall be provided with stud and terminal connectors near the place of attachment of the ROPS to the truck chassis. All floodlamps shall be adjustable in accordance with SAE J598. All lights shall be positioned within the outline of the truck.

3.26.2.1 Blackout lighting. A separate wiring harness may be provided for the blackout lights. One blackout headlight conforming to MS51318 shall be mounted on the extreme left within the plan outline of the forklift at the front, positioned to provide illumination with minimum obstruction when the forks are in carry position. The blackout headlight shall be adjustable in accordance with SAE J598. Two blackout stoplight-taillights conforming to MS51330 or MS52125 shall be mounted adjacent to the rear taillights, and shall be mounted in 6-inch-diameter holes or shall be provided with guards. Each

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light shall be recessed not less than 1/2 inch behind the hole or face of the guard. Two blackout marker lights conforming to MS52126 shall be mounted on the front, one each side of the vertical centerline of the truck, at the same level, and as far apart as practicable.

3.26.2.2 Taillights. Two taillights-stoplight assemblies in accordance with SAE J585 and SAE J586 shall be installed, one on the left rear and one on the right rear of the forklift, with the lens face recessed not less than 1/2 inch back of a protecting member.

3.26.3 Horn. The forklift shall be equipped with an electric, air, or air-over-electric horn. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions. The horn button shall be mounted on the steering wheel.

3.27 Personnel cab. When specified (see 6.2), the forklift shall be equipped with an insulated, weathertight personnel cab, which may be combined with ROPS and FOPS specified in 3.5.1. The cab clearances shall be in accordance with SAE J154. The cab shall withstand all operational shock and vibration without deformation or damage under the operating conditions specified herein. The cab shall be constructed of metal and shall be equipped with a minimum of one door and another emergency exit (either door or window that can be opened) on the other side of the cab. All doors shall have nonlocking closing latches, opening stops and hold-open catches. The doors shall either slide and latch at each end of travel or shall swing open a minimum of 90 degrees and latch open. All hardware shall be of the heavy duty truck or industrial type, resistant to corrosion. Front and rear window defrosters, operating in conjunction with the heater blower (see 3.27.4), or not less than two ventilation fans, not less than 200 cfm inclosed in a wire guard, shall be furnished. Fans shall be directionally adjustable to direct air flow toward all glazing. As a minumum, each fan shall have an "on-off" switch.

3.27.1 Glazing. As a minimum, glazing shall include a windshield, door windows and a rear window. Other windows, as necessary, shall be provided to give an upright seated operator all around visibility and visibility of the fork carrier in all possible fork positions. When a roof window is provided, it shall be sloped to provide for water drainage and guarded as necessary to meet the FOPS requirements (see 3.5.1). Glazing shall conform to ANSI Z26.1, type AS1 for the windshield and roof window and type AS1 or AS2 for other windows. Fixed or hinged glazing shall be sealed in waterproof, stress absorbent molding and movable windows shall be sealed on extra heavy type, pile line channels.

3.27.2 Interior. The interior of the cab shall be fitted with insulation retarding both heat loss and noise intrusion. Heavy-duty floor mats shall be furnished. All cab openings shall be sealed with elastomer grommets, seals, weather stripping, or body compound. The elastomer components shall withstand an ozone rating of two or better in accordance with ASTM D 1171.

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3.27.3 Windshield and window wipers. The cab shall be equipped with at least two powered windshield assemblies to clear the windshield and the rear window. The wiper motors shall provide not less than 90 inch-pounds of torque when electric motors are furnished, or provide not less than 170 inch-pounds of torque when air motors are furnished. Wipers shall be capable of variable stroke speeds up to at least 60 strokes per minute. Wiper arms and blades shall be of the heavy duty commercial type, having saddle-and-pin blade fastening. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. Air motors shall be plumbed to exhaust air outside of the cab. The operator controls shall be either installed on instrument panels or within easy reach of the operator within the cab.

3.27.4 Heater. A recirculating air type heater shall be furnished. The heater shall have a rating of at least 19,000 Btu/hour. The heater controls shall be within reach of the operator and shall provide for variable selection of blower speeds. If a fuel-fired heater is furnished, it shall be in accordance with SAE J1024.

3.28 Electromagnetic interference. The electromagnetic interference emission characteristics of the forklift shall meet the limits specified in MIL-STD-461, part 8, class C1, group II, except that a 20 dB relaxation is permitted in the emission limits.

3.29 Lubricants. Unless otherwise specified herein, the forklift shall be serviced with military lubricants selected by the contractor in accordance with table I, SAE J754. Run-in and testing shall be performed with these lubricants as applicable for the ambient operating temperature. A lubrication data plate identifying military lubricants and applicable temperature ranges shall be provided and mounted (see 3.32.1) on each forklift.

3.29.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with handoperated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with 3.7 for personnel normally clothed.

3.29.2 Enclosures with integral reservoir of lubricant. Enclosures such as the gearcases, transmission housing and engine crankcase, which contain a reservoir of liquid lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the acceptable level of the lubricant. As a minimum, dipsticks shall be graduated to show full and add levels. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drains, except planetaries, shall be fitted with a magnetic drainplug having either tapered pipe threads or shoulder and gasket. The drainplug shall be located so that removal of the plug will result in complete

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drainage of the lubricant from the enclosure. Drainage, except from planetary hubs, shall be to the ground when the forklift is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided without the removal or adjustment of accessories or parts. Plates equipped with hand operated fasteners may be removed.

3.30 Fungus and moisture resistance. Electrical circuitry, including all components and connections except as specified below, shall be protected from the effect of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition specified in 3.30.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish, or a varnish capable of providing the same protection and having the same or equal antipollution ingredients when required:

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.30.1 Composition I. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.31 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions that enable the forklift to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium shall be provided. Provisions shall conform to MIL-STD-209, classes 1 and 2 or class 3 for type II equipment, including the air transportation and helicopter external transport requirements. For forklifts with hinged-frame (articulated) steering, a steering frame lock in accordance with SAE J276 shall be provided except that the frame lock shall be painted in accordance with 3.33.

3.32 Identification marking. Each forklift shall be identified in accordance with MIL-STD-130 on identification plates conforming to MIL-P-514, type I, style 1, composition C (GG-P-455, type I, grade A, class 1) material. Plates shall be furnished and mounted by the contractor. A shipping data plate shall be furnished and shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material and, in addition, shall indicate the silhouette of the forklift in transport position indicating the center of gravity, the

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location and capacity of the slinging and the tiedown provisions. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.32.1 Instruction plates/labels. The forklift shall be equipped with instruction plates/labels conforming to MIL-P-514, type III, composition C (GG-P-455), type I, grade A, class 1) material, including warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating, towing, or servicing the forklift.

3.33 Treatment and painting. The portions of the forklift normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals and fiberglass, top coat color forest green conforming to MIL-C-46168. Parts and components previously primed by a subcontractor with alkyd base primer (color immaterial) may be used if cured for a minimum of 14 days and then cleaned and painted as specified herein either prior to or at final truck assembly painting. If another color is specified (see 6.2), the above system shall be overcoated with MIL-C-46168 of the desired color.

3.33.1 Brake protection. Surfaces of steel and iron brake assembly components which are exposed to air, except for the braking surface, shall be protected either by treatment conforming to MIL-T-704, type B, except that the finish coat is not required, or by a comparable commercial equivalent. Bearing surfaces in brake assemblies such as brake shoe anchor pins, anchor pin bearing holes in brake shoes, cylindrical surfaces of pistons in brake cylinders, and inside diameter of brake cylinders for the length of the piston contact shall be coated with a film of silicone compound conforming to MIL-C-21567. A film of silicone compound shall be applied to the lip and yoke hole of each dust boot and to the boot lip groove in the outside diameter at the ends of each cylinder. Protective treatment of brake components may be performed during initial component assembly.

3.34 Markings. Registration and other markings shall be applied in accordance with MIL-STD-642. The gross weight of the forklift shall be stenciled on each side of the forklift. The prescribed tire pressure shall be stenciled in a suitable location adjacent to each tire with block or stencil-type letters not more than 1 inch and not less than 3/4 inch. All suitable lifting tiedown points shall be marked either "Lifting", "Tiedown" or "Lifting and Tiedown". Unless otherwise specified (see 6.2), vehicle markings and their locations for Air Force use shall be in accordance with MIL-STD-1223. Registration numbers will be furnished by the contracting officer.

3.35 Toolbox. A toolbox shall be fabricated integrally inside the cab area or a separate toolbox made of steel having a minimum nominal thickness of 0.048 inch, shall be provided. The toolbox shall have a hinged lid and a hook fastener accessible and operable by personnel wearing gloves. The toolbox shall be bolted or welded to the forklift in a protected position, shall have provisions for drainage of liquids and shall accommodate any maintenance tools furnished. The toolbox shall be treated and painted in accordance with 3.33.

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3.36 Decontamination bracket assembly. A bracket assembly MS53052 will be furnished and mounted by the contractor. The location of the bracket shall be such that, it is accessible to personnel and provides clearance to remove and emplace a 5-gallon container conforming to MIL-C-1283.

3.37 Workmanship.

3.37.1 Castings and forgings. All parts, components, and assemblies of the forklift which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.37.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.37.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the forklift to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible, and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. The welding process used in fabrication of the forklift shall be at the option of the contractor.

3.37.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.
Welding Qualifications of the ASME, Section IX.
Earthmoving and Construction Equipment Welding, AWS/ANSI D14.3, Qualification, Section 5.

Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The certification shall be made available

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for review by the contracting officer or the contracting officer's representative.

3.37.3.2 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the forklift, the forklift shall conform to the inspection requirments specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.37.3.3 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and forklift.

3.37.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.37.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492 or IFI Standard No. 114 for Break Mandrel Blind Rivets.

3.37.6 Machine work. Tolerance and gages for metal fits shall conform to the limits specified herein and to the standards of the Materials Handling Equipment Industry or the Construction Equipment Industry.

3.38 Nuclear-biological-chemical (NBC) protection. When specified (see 6.2), the forklift shall be operable by personnel wearing the following complete protective ensemble:

a. Chemical-biological mask, MIL-M-51282.
b. Chemical-biological hood, MIL-H-51291.

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c. Chemical protective suit, MIL-S-43926.
d. Chemical protective glove set, MIL-G-43976.
e. Chemical protective footwear, MIL-F-43987.

The operator shall be capable of actuating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the forklift. The forklift shall complete two cycles in 9 minutes or less on the load placement course as specified in 4.6.2.41. The protective ensemble, as listed above, will be furnished as Government furnished equipment (GFE) in order to permit performance of the tests required under test method No. 36.

3.38.1 High altitude electromagnetic pulse (EMP). When specified (see 6.2), schematic and electrical diagram detailed to the component level (i.e., resistors, capacitors, filters, transistors, SCR's) with value of all components, of the forklift shall be furnished by the contractor to the Government to assist the Government in performing the EMP evaluation. A forklift shall be furnished to the Government and shall be evaluated by the Government to determine necessary protection devices for protection against EMP.

3.39 Vehicle weight classification sign kit. When specified (see 6.2), the contractor shall apply the Government furnished vehicle weight classification numbers to each vehicle utilizing a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle.

3.40 Government-furnished equipment (GFE). When specified (see 6.2), the following items shall be furnished by the Government (see 6.6).

a. Vehicle weight classification sign kit, NSN 9905-00-565-6267, for each truck.
b. Case, NSN 7520-00-559-9618 (the contractor shall furnish fasteners and install the case on the forklift).

3.40.1 Government-loaned equipment. When specified (see 6.2), the following items shall be loaned by the Government (see 6.6.1):

a. NBC protective ensemble for test method No. 36.

(1) Chemical-biological mask, NSN 4240-00-926-4200.
(2) Chemical-biological hood, NSN 4240-00-999-0420.
(3) Chemical protective suit, NSN 8415-00-407-1062.
(4) Chemical protective glove set, NSN 8415-01-033-3519.
(5) Chemical protective footwear, NSN 8430-01-021-5978.

b. STE/ICE for test in 4.6.2.43.1.
c. DCA tester for use in 4.6.2.43.1.
d. Cold-wet gloves.

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(1) Shell, NSN 8415-00-269-5700.
(2) Liner, NSN 8415-00-682-6575.

3.41 Diagnostic connector assembly (DCA) measurement capabilities. When specified (see 6.2), the forklift shall incorporate an easily accessible DCA in the operator's cab for interface with the Simplified Test Equipment/Internal Combustion Engine (STE/ICE) test equipment specified in MIL-T-62314, appendix B. The DCA shall be in accordance with TACOM drawing No. 12258941. All requirements for DCA shall be in accordance with the STE/ICE design guide for vehicle Diagnostic Connector Assemblies, report No. CR-82-588-003. The DCA shall have the capabilities for measuring the minimum functions defined in table II of report No. CR-82-588-003. A fuel shut-off method shall be provided for running compression unbalance checks. A separate wiring harness shall be provided for the DCA assembly and include all wiring and necessary hardware to perform required capabilities. The contractor shall also provide STE/ICE vehicle test cards as specified on TACOM drawing No. 12258955, and incorporate the test cards into forklift technical manuals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article inspection. Prior to examination and test of the forklift specified in 4.3.1 and 4.3.2, the following shall be performed:

a. Service the forklift with the oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
b. Conduct the run-in as prescribed by the contractor (see 4.6.2.38).
c. Set the hydraulic systems relief pressure as specified by the contractor and in 3.12 and record. Any change made to this setting shall constitute failure of all tests.

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d. Furnish a schedule of maintenance to be followed during all testing of the forklifts. A list of tools required to perform this maintenance shall also be provided.

4.3.1 First article examination.

4.3.1.1 Pretest examination. Prior to testing, one forklift shall be examinated for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test disassembly and examinations. Upon successful completion of all tests specified in 4.3.2, the truck shall be subjected to the tests marked "X" in column 2 of table III and 4.6.2.40. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. Upon successful completion of examination specified in 4.3.1.1, one preproduction forklift shall be subjected to the tests marked "X" in column 1 of table III except for the endurance tests (4.6.2.35). The other preproduction forklift shall be subjected to the endurance tests (see 4.6.2.35). Tests may be performed in any order. Except as otherwise specified herein, test conditions shall apply throughout testing. Failure of any test shall be cause for rejection. Failure of a preproduction forklift to meet any requirement specified herein shall be cause for rejection of the preproduction forklifts.

4.4 Inspection failure. Failure of a first article truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the first article truck(s), and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that correction action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all trucks accepted prior to the completion of first article inspection are similarly deficient, unless evidence to the contracy is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each forklift shall be examined for the defects marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each forklift shall be tested as specified in column 3 of table III. Failure of any test shall be cause for rejection.

4.5.3 Sampling.

4.5.3.1 Unit of product. A unit of product is defined as a fully-serviced forklift.

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4.5.3.2 Lot. A lot shall consist of not more than 25 units of product. Lots shall be formulated from forklifts which have successfully completed the individual examination and tests as specified in 4.5.1 and 4.5.2.

4.5.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the tests specified in 4.6.2.4, 4.6.2.5, 4.6.2.6, 4.6.2.17, and 4.6.2.18. Any sample failing any test shall be cause for rejection of the lot.

4.6 Inspection schedule.

4.6.1 Examination. Forklifts shall be examined as specified herein in accordance with table II.

TABLE II. Examination schedule.

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	101. Towbar or safety chains missing or not as specified.	3.1.3.1
X	-	102. Dissimilar metals not insulated as specified.	3.3.1
X	-	103. Screw threads not as specified.	3.3.2
X	X	104. Tire material not as specified.	3.3.3
X	X	105. Winterization provisions missing or not as specified.	3.4.5
X	X	106. Winterized engine enclosure not as specified.	3.4.5.2
X	X	107. Engine coolant heater not as specified.	3.4.5.3
X	X	108. Heater fuel supply system not as specified.	3.4.5.3.1
X	X	109. Heater exhaust system not as specified.	3.4.5.3.2
X	X	110. Winterized cab not as specified.	3.4.5.4
X	X	111. Cab insulation not as specified.	3.4.5.4.1
X	X	112. Weather stripping not as specified.	3.4.5.4.2
X	X	113. Cab heater and defroster not as specified.	3.4.5.5
X	X	114. Winterized battery enclosure not as specified.	3.4.5.6
X	X	115. Radiator winterfront not as specified.	3.4.5.7
X	X	116. Hose and hose assemblies not as specified.	3.4.5.8

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TABLE II. Examination schedule. (Continued)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	117. Safety provisions not as specified.	3.5
X	X	118. ROPS and FOPS missing or not as specified.	3.5.1
X	X	119. Seatbelt missing or not as specified.	3.5.2
X	X	120. Backup alarm missing or not as specified.	3.5.3
X	X	121. Human factors not as specified.	3.6
X	X	122. Maintainability provisions not as specified.	3.7
X	X	123. Calculated flotation index not as specified.	3.9.2
X	X	124. Engine not as specified.	3.10
X	X	125. Starting priming system missing or not as specified.	3.10.2.1
X	X	126. Alternator not as specified.	3.10.2.2
X	X	127. Starting system not as specified.	3.10.2.3
X	X	128. Fuel system not as specified.	3.10.2.4
X	X	129. Air cleaner not as specified.	3.10.2.5
X	X	130. Cooling system not as specified.	3.10.2.6
X	X	131. Fan not as specified.	3.10.2.6.1
X	X	132. Radiator not as specified.	3.10.2.6.2
X	X	133. Engine coolant hose not as specified.	3.10.2.6.3
X	X	134. Exhaust system not as specified.	3.10.2.8
X	X	135. Crankcase ventilation system not as specified.	3.10.2.9
X	X	136. Oil filters not as specified.	3.10.2.10
X	X	137. Drive assembly not as specified.	3.11
X	X	138. Powershift transmission not as specified.	3.11.1
X	X	139. Hydraulic system not as specified.	3.12
X	X	140. Hydraulic lines not as specified.	3.12.1
X	X	141. Pressure hose assemblies not as specified.	3.12.1.1
X	X	142. Pressure tubing not as specified.	3.12.1.2
X	X	143. Pressure tube fittings not as specified.	3.12.1.3
X	X	144. Suction line not as specified.	3.12.1.4
X	X	145. Return lines not as specified.	3.12.1.5
X	X	146. System filtration not as specified.	3.12.3 thru 3.12.3.2

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TABLE II. Examination schedule. (Continued)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	147. Reservoir not as specified.	3.12.4
X	X	148. Relief valve not as specified.	3.12.5
X	X	149. Hydraulic control valves not as specified.	3.12.6
X	X	150. Cylinders not as specified.	3.12.7
X	X	151. Hose reels not as specified.	3.12.10
X	X	152. Mast and fork carrier not as specified.	3.13
X	X	153. Wheels not as specified.	3.15.1
X	X	154. Tires not as specified.	3.15.2
X	X	155. Wheelguards not as specified.	3.16
X	X	156. Steering and steering control not as specified.	3.17
X	X	157. Service brakes not as specified.	3.18, 3.18.1
X	X	158. Parking brakes not as specified.	3.18, 3.18.2
X	X	159. Operator access not as specified.	3.19
X	X	160. Seat not as specified.	3.20
X	X	161. Hood not as specified.	3.21
X	X	162. Instrument panel not as specified.	3.22
X	X	163. Controls and instrumentation not as specified.	3.23
X	X	164. Light switches missing or not as specified.	3.23.1
X	X	165. Engine run control not as specified.	3.23.2
X	X	166. Starter system control not as specified.	3.23.3
X	X	167. Braking and inching controls not as specified.	3.23.4
X	X	168. Accelerator control not as specified.	3.23.5
X	X	169. Transmission controls not as specified.	3.23.6
X	X	170. Hydraulic controls not as specified.	3.23.7
X	X	171. Electrical system monitor missing or not as specified.	3.23.8
X	X	172. Engine oil pressure monitors missing or not as specified.	3.23.9
X	X	173. Engine coolant temperature monitor missing or not as specified.	3.23.10
X	X	174. Transmission temperature monitor missing or not as specified.	3.23.11
X	X	175. Hourmeter missing or not as specified.	3.23.12

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TABLE II. Examination schedule. (Continued)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	176. Fuel gage missing or not as specified.	3.23.13
X	X	177. Air pressure monitor missing or not as specified.	3.23.14
X	X	178. Batteries not as specified.	3.24.1
X	X	179. Battery mounting not as specified.	3.24.2
X	X	180. Battery cable terminals not as specified.	3.24.3
X	X	181. Slave receptacles missing or not as specified.	3.24.4
X	X	182. Pintle hook missing or not as specified.	3.25
X	X	183. Circuit breakers missing or not as specified.	3.26.1
X	X	184. Lights missing or not as specified.	3.26.2
X	X	185. Blackout lighting missing or not as specified.	3.26.2.1
X	X	186. Taillights missing or not as specified.	3.26.2.2
X	X	187. Horn missing or not as specified.	3.26.3
X	X	188. Personnel cab not as specified.	3.27
X	X	189. Glazing not as specified.	3.27.1
X	X	190. Cab interior not as specified.	3.27.2
X	X	191. Windshield and window wipers not as specified.	3.27.3
X	X	192. Heater missing or not as specified.	3.27.4
X	X	193. Lubrication not as specified.	3.29
X	X	194. Lubrication fittings not as specified.	3.29.1
X	X	195. Filling, draining, and checking provisions for lubricant enclosures missing or not as specified.	3.29.2
X	X	196. Fungus and moisture resistance provisions not as specified.	3.30, 3.30.1
X	X	197. Slinging and tiedown provisions (and frame lock if applicable) missing or not as specified.	3.31
X	X	198. Identification marking missing or not as specified.	3.32
X	X	199. Instruction plates missing or not as specified.	3.32.1

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TABLE II. Examination schedule. (Continued)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	101a. Treatment and painting not as specified.	3.33
X	X	102a. Brake protection not as specified.	3.33.1
X	X	103a. Marking not as specified.	3.34
X	X	104a. Castings and forgings not as specified.	3.37
X	X	105a. Metal fabrication not as specified.	3.37.2
X	X	106a. Welding not as specified.	3.37.3 thru 3.37.3.3
X	X	107a. Welding certification not as specified.	3.37.3.1
X	X	108a. Bolted connections not as specified.	3.37.4
X	X	109a. Riveted connections not as specified.	3.37.5
X	X	110a. Machine work not as specified.	3.37.6
X	X	111a. Parts or components missing or do not function.	Section 3
X	X	112a. Assembly incorrect or not complete.	Section 3
X	X	113a. Materials not as specified.	3.3
X	X	114a. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	3.3.1.1
X	X	115a. Drain holes were not provided.	3.3.1.2
X	X	116a. Contractor does not have documentation available for identification of material, material finishes or treatments.	3.3.1.3
X	X	117a. Tool box not as specified.	3.3.5
X	X	118a. Decontamination bracket assembly not as specified.	3.36
X	X	119a. Vehicle weight classification sign kit not as specified.	3.39

4.6.2 Tests. Test shall be conducted in accordance with table III. An "X" in the column indicates the tests that shall be conducted.

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4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. All tests shall be conducted with ROPS and FOPS installed. Safety and human factor considerations (see 3.5 and 3.6), shall be evaluated throughout testing. Test methods referenced in tests specified herein are those contained in the appendix. Operation of the forklift shall be with fuels, oils, lubricants, and antifreeze specified herein as applicable for the temperature conditions existing at place of test. All tests shall be conducted at the contractor's recommended tire pressure.

TABLE III. Test schedule.

First article	Post-test	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	X	Test conditions.	4.6.2.1	3.5, 3.6, 3.29
X	-	X	Engine.	4.6.2.2	3.10
X	-	-	Weight and dimensions.	4.6.2.3	3.1.2.1 thru 3.1.2.4, 3.9.11 thru 3.9.14, 3.15.2
X	X	-	Lifting speed and lowering.	4.6.2.4	3.9.7, 3.9.8
X	-	-	Curb clearance circle.	4.6.2.5	3.9.9
X	X	-	Travel speed.	4.6.2.6	3.9.4, 3.10.2.8, 3.11.1
X	-	-	Underclearance, angles of approach and departure.	4.6.2.7	3.1.2.5
X	-	-	Upright tilt, tilt speed, static stability, tilt cavitation, mast clearance.	4.6.2.8	3.13, 3.9.1.1, 3.9.10
X	X	-	Drift.	4.6.2.9	3.9.15
X	-	-	Longitudinal gradeability.	4.6.2.10	3.9.3, 3.10.2.8
X	-	-	Dynamic, stability, parking brake, emergency brake.	4.6.2.11	3.9.1.2, 3.18.2, 3.18.3
			Steering.	4.6.2.12	-
X	-	-	Obstacle course No. 1.	4.6.2.12.1	3.17
X	-	-	Obstacle course No. 2.	4.6.2.12.2	3.17
X	-	-	Emergency steering.	4.6.2.12.3	3.17.2
X	-	-	Steering control.	4.6.2.12.4	3.17.1

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TABLE III. Test schedule. (Continued)

First article	Post-test	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Highway & rail transportability.	4.6.2.13	3.1.3.1, 3.1.3.3
X	-	-	ROPS and FOPS.	4.6.2.14	3.5.1
X	-	-	Overload.	4.6.2.15	3.14
X	-	-	Torque converter.	4.6.2.16	3.11.1, 3.25
X	-	-	Inching/transmission disconnect.	4.6.2.17	3.11, 3.11.1, 3.12, 3.23.4, 3.10.2.6, 3.10
X	-	-	Service brake.	4.6.2.18	3.18.1
X	-	-	Maintenance.	4.6.2.19	3.4.5.2, 3.4.5.3, 3.4.5.3.2, 3.7
X	-	-	Air transportability.	4.6.2.20	3.1.3.2
			Hydraulic system.	4.6.2.21	-
X	-	-	Pump inlet pressure.	4.6.2.21.1	3.12.1.4
X	-	-	Hydraulic filter and filter assembly.	4.6.2.21.2	3.12.3, 3.12.3.1, 3.12.3.2
X	X	X	Hydraulic system cleanliness.	4.6.2.21.3	3.12.9
X	-	X	Hydraulic cylinders.	4.6.2.21.4	3.12.7
X	-	-	Pump contaminent tolerance.	4.6.2.21.5	3.12.2
X	-	-	Hydraulic fittings.	4.6.2.21.6	3.12.1.1
X	-	-	Container operations.	4.6.2.22	3.6, 3.9.17
X	-	-	Sideshift, oscillation, and fork spread.	4.6.2.23	3.13.1, 3.9.16, 3.14
X	-	-	Electromagnetic interference.	4.6.2.24	3.28
X	-	-	Slinging provisions.	4.6.2.25	3.31
X	-	-	Tiedown provisions.	4.6.2.26	3.31
			Noise level measurement.	4.6.2.27	-
X	-	-	Operator's station.	4.6.2.27.1	3.6.1
X	-	-	Exterior spectator.	4.6.2.27.2	3.6.1
X	-	-	Noise hazard.	4.6.2.27.3	3.6.2
X	-	-	Rain.	4.6.2.28	3.10.2.8, 3.4.4
X	-	-	Wheel guards.	4.6.2.29	3.16
X	-	-	Low temperature (-25° F).	4.6.2.30	3.4.2, 3.4.3, 3.10, 3.10.2.6, 3.12
X	-	-	Fording.	4.6.2.31	3.9, 3.9.5, 3.10.2.3, 3.10.2.9

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TABLE III. Test schedule. (Continued)

First article	Post-test	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
-	-	-	Low temperature winterization (-50° F).	4.6.2.32	3.4.3, 3.4.5 thru 3.4.5.7
X	-	-	High speed.	4.6.2.33	3.9.19
X	-	-	Controls.	4.6.2.34	3.23.6, 3.23.7
X	-	-	Endurance.	4.6.2.35	3.9.18, 3.10.2.4
-	-	-	Reliability.	4.6.2.36	3.8, 3.9.6, 3.10, 3.10.2.4, 3.10.2.6, 3.11, 3.12
-	-	-	Maintainability.	4.6.2.37	3.7
X	-	X	Run-in.	4.6.2.38	Section 3
X	-	-	Rear axle oscillation.	4.6.2.39	3.11.2
X	X	-	Post-test.	4.6.2.40	Section 3

4.6.2.2 Engine. The diesel engine shall be tested or certified in accordance with 5.2 and 5.3 of MIL-STD-1410.

4.6.2.3 Weight and dimensions. Determine the weight and dimensions in accordance with test method No. 1. Nonconformance to 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4, 3.9.11, 3.9.12, 3.9.13, 3.9.14, or 3.15.2 shall constitute failure of this test.

4.6.2.4 Lifting speed and lowering speed. Test the forklift in accordance with test method No. 2. Nonconformance to 3.9.7 or 3.9.8 shall constitute failure of this test.

4.6.2.5 Curb clearance circle. Determine the curb clearance circle in forward and reverse as defined in Field Test Procedure of SAE J695 with rated load in carry position. Nonconformance to 3.9.9 shall constitute failure of this test.

4.6.2.6 Travel speed. Test the forklift in accordance with test method No. 3. Speeds not as specified in 3.9.4 shall constitute failure of this test. Inability to shift as specified in 3.11.1 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

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4.6.2.7 Underclearance, angles of approach, and departure. Test the forklift in accordance with test method No. 4. Nonconformance to 3.1.2.5 shall constitute failure of this test.

4.6.2.8 Upright tilt, tilt speed, static stability, tilt cavitation, mast clearance. Test the forklift in accordance with test method No. 5. Nonconformance to 3.9.1.1, 3.9.10, or 3.13 shall constitute failure of this test.

4.6.2.9 Drift. Test the forklift in accordance with test method No. 6. Nonconformance to 3.9.15 shall constitute failure of this test.

4.6.2.10 Longitudinal gradeability. Test the forklift in accordance with test method No. 7. Inability of the forklift to ascend the slope as specified in 3.9.3 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.11 Dynamic stability, parking brake, emergency brake. Test the forklift in accordance with test method No. 8. Nonconformance to 3.9.1.2, 3.18.2, or 3.18.3 shall constitute failure of this test.

4.6.2.12 Steering.

4.6.2.12.1 Obstacle course No. 1. Test the forklift in accordance with test method No. 9, 3a and 3b. Inability of the forklift to negotiate the test course or the hitting of any two markers for any run shall constitute loss of operator control specified in 3.17 and shall constitute failure of this test.

4.6.2.12.2 Obstacle course No. 2. Test the forklift in accordance with test method No. 9, 3c. Inability of the forklift to negotiate the test course or hitting of any markers shall constitute loss of operator control specified in 3.17 and shall constitute failure of this test.

4.6.2.12.3 Emergency steering. Test the emergency steering capability of the forklift in accordance with test method No. 10. Inability of the forklift to negotiate the test course or failure to contact all markers on the test course shall constitute inability to retain steering ability specified in 3.17.2 and shall constitute failure of this test.

4.6.2.12.4 Steering control. Test the steering control in accordance with test method No. 11. Nonconformance to 3.17.1 shall constitute failure of this test.

4.6.2.13 Highway and rail transportability. Test the forklift in accordance with test method No. 12. Nonconformance to 3.1.3.1 or 3.1.3.3 shall constitute failure of this test.

4.6.2.14 ROPS and FOPS. The ROPS and FOPS shall be tested in accordance with SAE J1040 and SAE J1043. Nonconformance to 3.5.1 shall constitute failure of this test.

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4.6.2.15 Overload. Test the forklift in accordance with test method No. 13. Nonconformance to 3.14 shall constitute failure of this test.

4.6.2.16 Torque converter. Test the forklift in accordance with test method No. 14. Nonconformance to 3.11.1 or 3.25 shall constitute failure of this test.

4.6.2.17 Inching/transmission disconnect. Test the forklift in accordance with test method No. 15. Nonconformance to 3.11.1 or 3.23.4, or exceeding the temperature limits specified for the cooling systems (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11), or hydraulic system (see 3.12), shall constitute failure of this test.

4.6.2.18 Service brake. Test the forklift in accordance with test method Nos. 16, 17, and 18. Nonconformance to 3.18.1 shall constitute failure of this test. For initial production and sample test only, test method No. 16 is required for braking.

4.6.2.19 Maintenance. Test the forklift in accordance with test method No. 19. Nonconformance to 3.7, 3.4.5.2, 3.4.5.3, or 3.4.5.3.2 as applicable shall constitute failure of this test.

4.6.2.20 Air transportability. Perform examination and tests required by MIL-A-8421. Nonconformance to the requirements of 3.1.3.2 shall constitute failure of this test.

4.6.2.21 Hydraulic system.

4.6.2.21.1 Pump inlet pressure. Test each pump in accordance with test method No. 20. Pressure differential between atmospheric pressure and the vacuum in the line in excess of that specified in 3.12.1.4 shall constitute failure of this test.

4.6.2.21.2 Hydraulic filter and filter assembly. Test the hydraulic filter in accordance with test method No. 21. Nonconformance to 3.12.3, 3.12.3.1, or 3.12.3.2, as applicable, shall constitute failure of this test.

4.6.2.21.3 Hydraulic system cleanliness. Test each truck in accordance with test method No. 22. Continue cleanup procedures until the average of not less than three consecutive counts conform to 3.12.9.

4.6.2.21.4 Hydraulic cylinders. Test each cylinder in accordance with test method No. 23. Nonconformance to 3.12.7 shall constitute failure of this test.

4.6.2.21.5 Pump contaminant tolerance. Test the pump(s) in accordance with procedure set up in NFPA Recommended Standard T.3.9.18-1976. Using the test results with the computer program furnished by the procuring activity (see 6.7), calculate and present a 1000 hour Contaminant Tolerance Profile on the particulate contamination chart shown on figure 2. The profile nonconformance to 3.12.2 shall constitute failure of this test. Report contamination analysis data in accordance with ANSI Standard B93.30M (see 3.12.2).

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4.6.2.21.6 Hydraulic fittings. Test the truck in accordance with test method No. 34. The contractor shall furnish a complete set of hose assemblies with reusable fittings for conduct of this test. Nonconformance to 3.11.1.1 shall constitute failure of this test.

4.6.2.22 Container operations. Test the forklift in accordance with test method No. 24. Nonconformance to 3.6 or 3.9.17 shall constitute failure of this test.

4.6.2.23 Sideshift, oscillation, and fork spread. Test the forklift in accordance with test method No. 25. Nonconformance to 3.13.1, 3.9.16 or 3.14 shall constitute failure of this test.

4.6.2.24 Electromagnetic interference. The forklift shall be tested in accordance with MIL-STD-461. Nonconformance to 3.28 shall constitute failure of this test.

4.6.2.25 Slinging provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD-209. Nonconformance to 3.31 shall constitute failure of this test.

4.6.2.26 Tiedown provisions. Test the tiedown provisions by applying the ultimate load for the time period specified in MIL-STD-209. Nonconformance to 3.31 shall constitute failure of this test.

4.6.2.27 Noise level measurement.

4.6.2.27.1 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, paragraph 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position, additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6.2.27.2 Exterior spectator. Measure the exterior noise level in accordance with SAE J88. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.2.27.3 Noise hazard. Measure the noise level at the operator station and on a 5-foot perimeter while operating the truck at maximum governed speed raising rated load. If the noise level exceeds 85dB(A) at any location, noise hazard warning signs and the appropriate discussion of noise hazard specified in 3.6.2 shall be provided.

4.6.2.28 Rain. Test the forklift in accordance with test method No. 26. Nonconformance to 3.4.4 or 3.10.2.8 shall constitute failure of this test.

4.6.2.29 Wheel guards. Apply a vertical downward force of 300 pounds to the geometric center of the horizontal surface, or to the uppermost point (not including supports or braces), whichever is applicable. The force shall be

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applied for not less than 2 minutes. Examine the guard for permanent deformation for fractures. Flexible wheel guards may be deformed during the test loading but shall return to their original shape when the load is removed. Nonconformance to 3.16 shall constitute failure of this test.

4.6.2.30 Low temperature (-25° F). Test the forklift in accordance with test method No. 27. One or more of the following shall constitute failure of this test:

a. Damage to any part of the forklift due to storage at -30° F (see 3.4.3).
b. Engine does not start within 5 minutes after beginning the first cranking (see 3.4.2).
c. Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of primers or priming aids (see 3.4.2).
d. Instruments and gages do not operate (see 3.4.2).
e. Battery charging system does not show positive charge (see 3.4.2).
f. With the operator wearing arctic mittens and boots:

    (1) Power train cannot be engaged to the wheels (see 3.4.2).
    (2) Any clutch, brake, shaft, or cylinder cannot be operated through two complete movement cycles (see 3.4.2).

g. Operator encounters difficulty in mounting to and dismounting from the operator's position on the forklift (see 3.4.2).
h. Measured temperatures of engine oil (see 3.10), of engine coolant (see 3.10.2.6), or hydraulic oil (see 3.12) not within specified limits.

4.6.2.31 Fording. The forklift shall be tested in accordance with test method No. 28. Any of the following shall constitute failure of this test:

a. Evidence of water damage or malfunction of the torque converter, engine oil sump, transmission, transfer case, fuel tank, hydraulic reservoir, either axle, any wheel, instruments, or gages (see 3.9.5).
b. Malfunction of any electrical component (see 3.9.5).
c. Malfunctioning of any brake (see 3.9.5).
d. Malfunctioning of any hydraulic component (see 3.9.5).
e. Malfunctioning of the steering system (see 3.9.5).
f. Evidence of corrosion which affects the correct functioning of the forklift (see 3.9.5).
g. Water entering the engine (see 3.10.2.9).
h. Inability to start the engine in the water (see 3.10.2.3).
i. Volumetric water content in excess of that specified (see 3.9.5).

4.6.2.32 Low temperature winterization (-50° F). Test the forklift in accordance with test method No. 29. Nonconformance to 3.4.3 or 3.4.5 through 3.4.5.7 shall constitute failure of this test.

4.6.2.33 High speed. Test the forklift in accordance with test method No. 30. Nonconformance to 3.9.19 shall constitute failure of this test.

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4.6.2.34 Controls. Test the controls in accordance with test method No. 33. Nonconformance to 3.23.6 or 3.23.7 shall constitute failure of this test.

4.6.2.35 Endurance. The forklift shall be tested in accordance with test method Nos. 31 and 32. Test method No. 31 shall be conducted for not less than 4000 cycles. Test method No. 32 shall be conducted for not less than 250 cycles in not more than 300 hours. Nonconformance to 3.9.18 or 3.10.2.4 shall constitute failure of this test.

4.6.2.36 Reliability. Using the MTBF specified in 3.8, the forklift(s) shall be tested in accordance with test method No. 32 for an "accept" or "reject" decision in accordance with MIL-STD-781, test plan IV. A "reject" decision shall constitute failure of this test. For the purpose of determining MTBF, a failure is defined as any malfunction which causes any unscheduled maintenance action to be taken in order to restore the forklift to specified requirements. Those malfunctions resulting from accidents, noncompliance with prescribed operational or maintenance procedures, faulty test instrumentation, fail-safe devices operating as specified, and dependent secondary failures shall be considered nonrelevent failures. Nonrepetitive replacement of light bulbs, fuses, tires, and tightening of fasteners shall be considered nonrelevent failures. Any impending malfunction detected which constitutes a safety hazard to operating personnel or would cause secondary damage either to the item under test or testing instrumentation will be corrected prior to malfunction and considered as a relevent failure. Hours specified herein shall be productive clock hours excluding fueling, servicing, and downtime. In addition, failure to meet the fuel tank capacity specified in 3.10.2.4, the load transfer rate specified in 3.9.6, or exceeding the temperature limits specified for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11), universal joints (see 3.11), or hydraulic system (see 3.12) shall constitute failure of this test.

4.6.2.37 Maintainability. Maintenance times shall be recorded throughout testing in accordance with 4.6.2.36. A maintenance ratio of more than 0.15 as specified in 3.7 shall constitute failure of this test.

4.6.2.38 Run-in. Each forklift shall be driven through a run-in for not less than 1 hour. The run-in shall include operating in all gears, operating all electrical equipment, and performing all hydraulic functions with rated load as well as no load. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any component, or evidence of leakage of air fuel, lubricant or hydraulic oil, shall be cause for rejection of the forklift by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

4.6.2.39 Rear axle oscillation. Measure the total rear axle oscillation in accordance with test method No. 35. Nonconformance to 3.11.2 shall constitute failure of this test.

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4.6.2.40 Posttest disassembly and examination. Upon completion of all tests, measure and record the compression in each engine cylinder and dismantle the truck sufficiently to permit detailed disassembly and visual examination of the following components:

a. Engine cylinder heads, all valves, two pistons and piston pins taken from the two cylinders having the lowest compression readings at the end of the test, one main bearing and one connecting rod bearing.
b. Torque converter and transmission.
c. Universal joints and drive shafts.
d. Differential.
e. Front axle bearings and seals.
f. Complete brake system components.
g. Steering system cylinders, valves, and pump.
h. Hydraulic system pump, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the test.
i. Steer axle bearings and seals.
j. Mast assembly bearings and chains.

Additional disassembly and examination shall be made at the option of the contracting officer. The truck shall be examined in accordance with table II, column 1. Record complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Each evidence of corrosion which affects the functions of the truck or evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.6.2.41 Nuclear-biological-chemical (NBC) protection. The forklift shall be tested in accordance with test method No. 36. Nonconformance to 3.38 shall constitute failure of this test.

4.6.2.42 Electromagnetic pulse (EMP). The Government will evaluate the schematics and electrical diagrams to determine possible problem areas that may be encountered on the end item. A third preproduction forklift shall be subjected to non-destructive EMP testing at a Government facility.

4.6.2.43 DCA capability. The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation of the forklift to insure wire and connection are free of error. After installation on the forklift, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.41 shall constitute failure of the DCA.

4.6.2.43.1 DCA - STE/ICE interfaces. Upon completion of inspection and tests specified in 4.6.2.43, test the DCA for all measurements specified in 3.41 using STE/ICE. Nonconformance to 3.41 shall constitute failure of DCA.

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4.7 Inspection comparison. The Government may select forklifts at any time during the contract production period and subject these forklifts to the examination specified in 4.6.1 and column 1 of table II, and to the tests marked "X" in column 1 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article forklifts. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison forklift to meet any requirement specified herein, during and as a result of the examination and test specified in 4.7, shall be cause for rejection of the inspection comparison forklift(s), and shall be cause for refusal by the Government to continue acceptance of production forklifts until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on forklifts previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all forklifts accepted prior to the completion of inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor, and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of the referenced documents in section 5 of this specification, and to determine compliance with the specific commercial preservation, packing and marking requirements.

5. PACKAGING

5.1 Preservation, packing and marking. Each forklift truck shall be preserved, packed and marked in accordance with the requirements of MIL-STD-162 for type I mobile shipment. For all degrees of preservation, liquid cooling systems shall be preserved by filling the system with equal parts of water and antifreeze conforming to MIL-A-46153. The degree of preservation shall be level A, B, C, or commercial; and the degree of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The forklift is intended for use as a materials handling forklift with a capability of stuffing and stripping the 8 foot wide x 8 foot high family of ANSI/ISO cargo containers, transporting, and loading boxes, containers, and palletized loads up to 4000 pounds. The mobility of the forklift, with and without payload, is intended to cover working areas where high flotation and traction characteristics are required.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article tests are not required (see 3.2.1).
c. Time frame required for submission of the first article model and number of forklifts required (see 3.2.1).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify time frame and which examination and tests will be conducted by the Government, and which examination and tests shall be conducted by the contractor (see 3.2.1).
e. When winterized forklift is required (see 3.4.5 and 3.2.1).
f. When a personnel cab is required (see 3.27).
g. When vehicles procured for Air Force shall be marked other than in accordance with MIL-STD-1223 (see 3.34).
h. Oil sampling valve on the transmission if required (see 3.10.2.12).
i. Degree of preservation and packing required (see 5.1).
j. When nuclear-biological-chemical (NBC) protection is not required (see 3.38).
k. When high altitude electromagnetic pulse (EMP) is not required (see 3.38.1).
l. When weight classification sign kit is not required (see 3.39).
m. When Government-furnished equipment (GFE) is not required (see 3.40).
n. When Government loaned equipment is not required (see 3.40.1).
o. When DCA is not required (see 3.41).

6.3 First article. When a first article inspection is required, the items should be an initial production model. The first article should consist of two or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.3.1 Incident report. The contracting officer should require the contractor to furnish a written report within 24 hours of any incident of equipment malfunction or failure, during the conduct of any test. As a minimum, the report should describe components and parts affected, test and operating conditions, date of incident, hourmeter reading, how detected, and description of incident.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.5 Noise. When the noise level of the forklift exceeds 85dB(A) the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS Paragraph) should include the requirement for hearing protection, the noise level of the forklift, the distance from the forklift at which 85dB(A) will always be met, and when operational situations or system configuration will reduce the level to 85dB(A).

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6.6 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.40.

6.6.1 Government-loaned equipment. The contracting officer should arrange to loan the equipment specified in 3.40.1.

6.7 Contaminant tolerance profile. The contracting officer should arrange to furnish the computer program for Contaminant Tolerance Profile required in 4.6.2.21.5, when requested by the contractor.

6.8 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the forklifts covered by this specification, as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.9 Definitions.

6.9.1 Load center. The horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load engaging faces of the forks to the center of gravity of the load.

6.9.2 Rated capacity load. Rated capacity load shall be established with a load equivalent to an unrestricted cube with overall dimensions twice the load center dimensions, and whose center of gravity is located at the geometrical center of the cube.

6.9.3 Carry position. Carry position is defined as the distance from the heel of the forks to the ground with the forks tilted fully to the rear with load centered and shall be 12 inches.

6.10 Government furnished equipment. The contracting officer should arrange to furnish the mobile ramp for container operations required in 4.6.2.22, 4.6.2.35 and 4.6.2.36.

6.11 Spare parts. The contracting officer shall include provisioning requirements for spare parts.

Custodians:
Army - ME
Navy - SA
Air Force 99

Review activities:
DLA - CS
Army - AT
Navy - OS
Air Force - 84

User activities:
Navy - MC

Preparing activity:
Army - ME

Project 3930-0527

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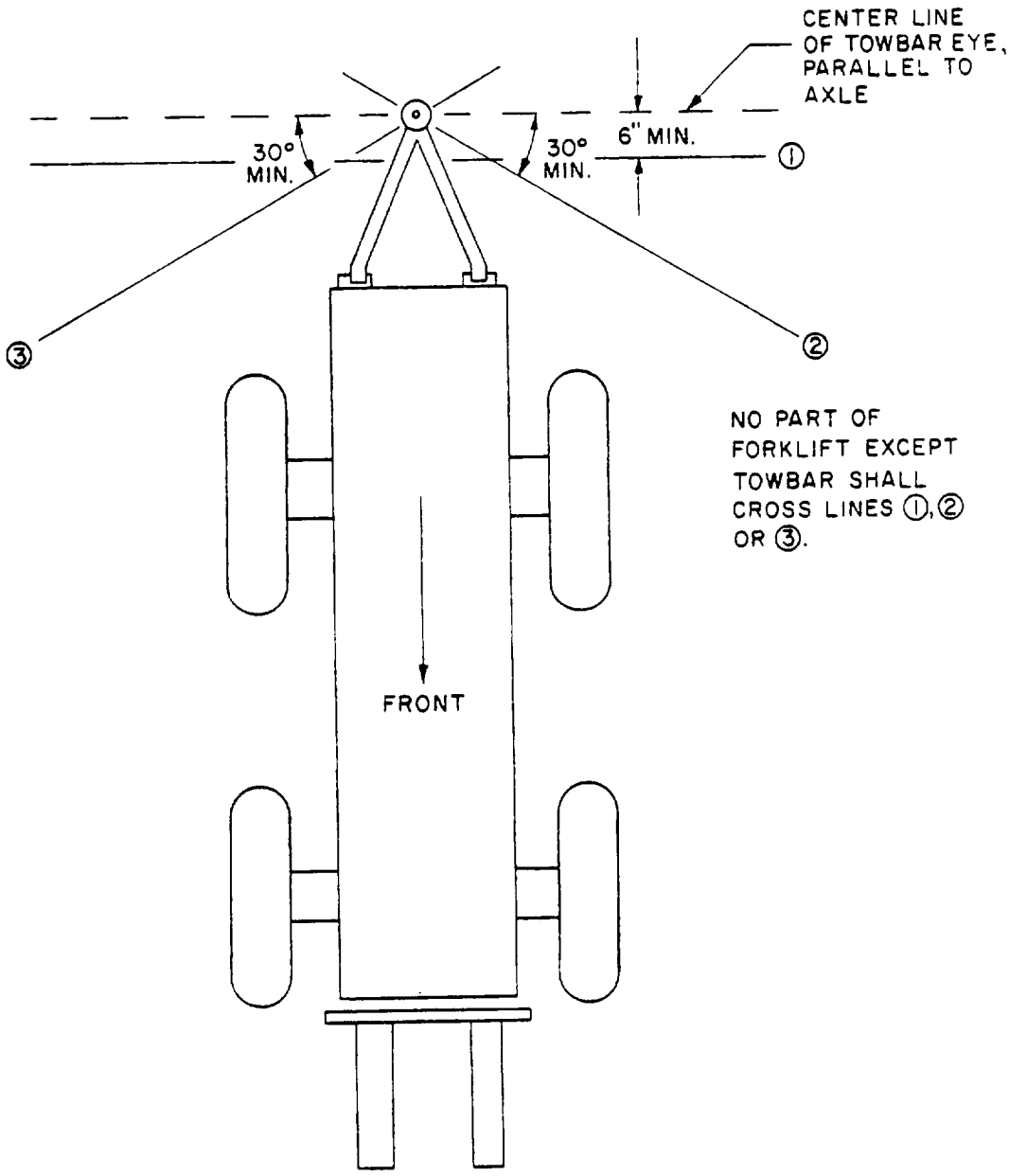


FIGURE 1. TOWBAR TURNING CLEARANCE AREA

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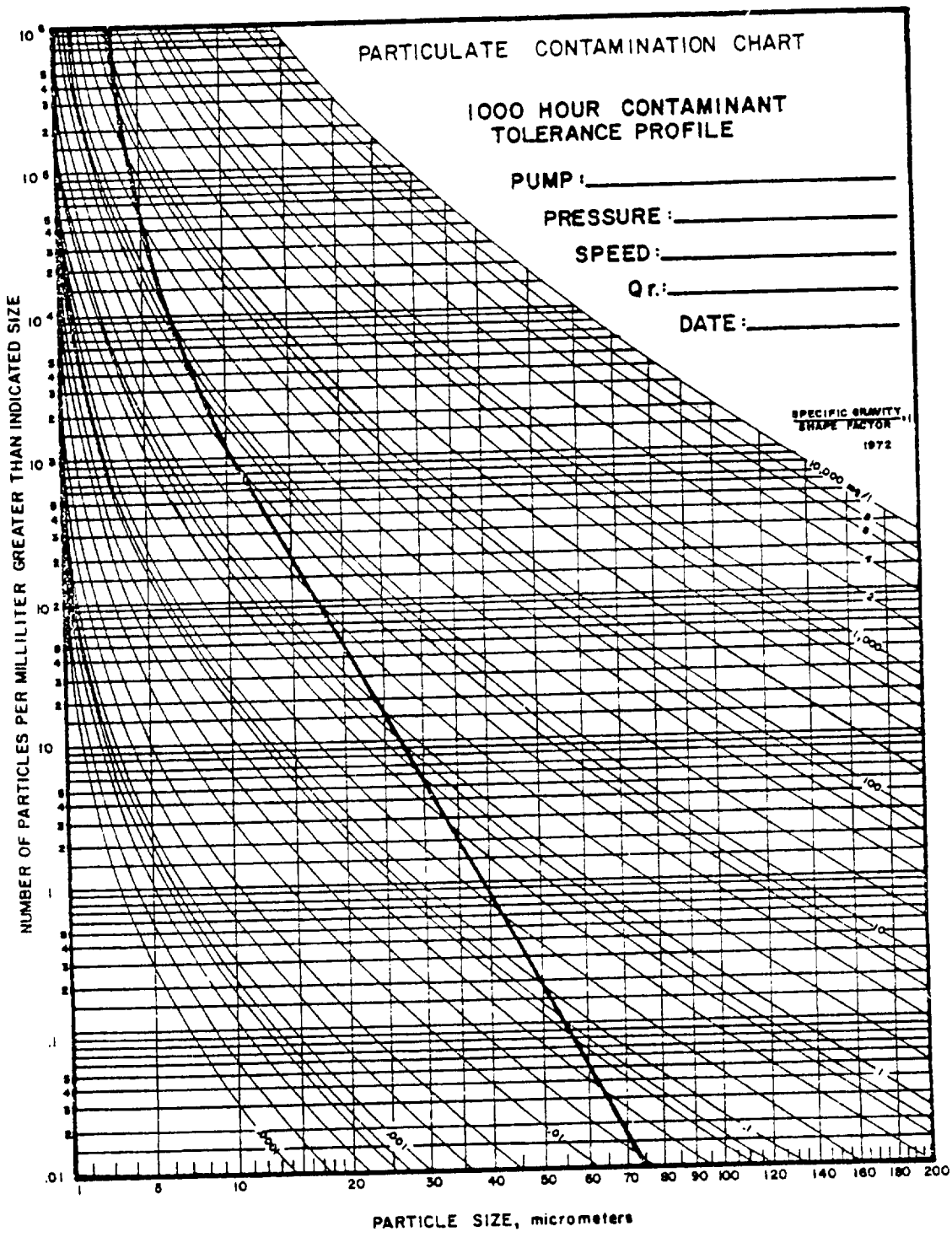


FIGURE 1. CHART FOR CONTAMINANT TOLERANCE PROFILE.

X-2582

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APPENDIX

FOR

TRUCKS, LIFT, FORK, DIESEL-ENGINE-DRIVEN:

PNEUMATIC-TIRED, ROUGH TERRAIN

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether forklifts procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B93.28 - Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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TEST METHOD NO. 1

WEIGHT AND DIMENSIONS

1. TEST COURSE:

   a. Level, flat surface.
   b. Level, flat surface with sharp drop-off over an edge.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Liquid level or clinometer.

3. TEST PROCEDURE:

   a. Lift the rated load to the full extent of mast travel. Adjust the fork tilt to level the load. Measure the true vertical distance from the surface to the top surface of the forks. (This is the maximum fork height.)
   b. With mast vertical, raise the unloaded forks until the inner mast begins to exceed the specified collapsed mast height. Measure and record the true vertical distance to the top surface of the forks at the specified load center. (This is the lift height.)
   c. Measure the true vertical distance from the floor to the uppermost projection of the upright assembly with mast in true vertical position without load as determined by a clinometer. (This is the collapsed mast height.)
   d. Position the forklift on the edge of the drop-off. Lower the forks (without load) over the edge to the full extent of boom travel. Adjust the fork tilt to level the top surface of the forks. Measure the true vertical distance between the surface on which the forklift is positioned and the top surface of the forks. (This is the low engagement height.)
   e. With collapsed mast in vertical position, measure the height of the unloaded truck in accordance with SAE J894.
   f. With carriage centered, measure the width of the unloaded truck in accordance with SAE J894.
   g. With the forks removed, measure the length of the unloaded truck in accordance with SAE J894.
   h. Weigh the forklift fully fueled and serviced, without load. Obtain front and rear axle weights with and without rated load in carry position (see 6.9.3), for use in caclulating the Flotation Index. The forklift shall be in the straight ahead position.

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TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch.

2. TEST PROCEDURE:

   a. Record time in seconds required to raise rated load from ground to maximum lift height (with mast vertical) at maximum engine speed.
   b. Record time to lower rated load at maximum speed (full open valve) to 3 to 4 foot height. Load to be abruptly stopped at the 3 to 4 foot height. Record height at stop. Engine to be at low idle speed.
   c. Repeat steps "a" and "b" a minimum of 3 times.
   d. Compute lifting speed and lowering speed and use the fastest rate obtained from each as the established speeds.
   e. Remove load and raise empty forks to maximum fork height. Record time in seconds required to lower forks to lowered position. Repeat 3 times.
   f. Compute the lowering speed and use the fastest rate as the established speed.

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TEST METHOD NO. 3

TRAVEL SPEED

1. TEST COURSE:

   a. The test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, plus efficient stopping distance.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch or electronic timer.
   c. Pressure measurement device.

3. TEST PROCEDURE:

   a. Install exhaust system pressure measurement device not more than 2 inches beyond the header outlet. Record exhaust pressure for each run.
   b. Drive the forklift to attain maximum speed prior to entering the measured test course. (Do not exceed an operating speed at which the forklift cannot be controlled.) While accelerating, perform gear or range changes under full throttle. Forks shall be in the carry position, (full carriage tilt backward).
   c. Drive through the measured distance at maximum speed, recording the length of the distance and the time required to traverse it. The measured distance shall be not less than 88 feet.
   d. Turn the forklift around and repeat the test approaching the measured distance from the opposite end. Repeat for a total of six runs, three in each direction through the measured distance.
   e. Conduct procedure 2a through 2d for the following:

      (1) Forward with rated load.
      (2) Rearward with rated laod.
      (3) Forward without load.
      (4) Rearward without load.

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TEST METHOD NO. 4

UNDERCLEARANCE, ANGLES OF APPROACH AND DEPARTURE

1. TEST COURSE:

   a. Level, flat surface.
   b. Ramp inclined to 20 degrees and a length greater than 1.2 times the truck wheelbase. At the top shall be a horizontal surface forming a sharp angle with the ramp.

2. TEST APPARATUS:

   a. Linear measurement equipment.

3. TEST PROCEDURE:

   a. Test course 1a above. Tires shall be inflated to the pressure indicated in the TRA Yearbook for maximum tire loading at 5 mph rating. Raise the rated load to carry position, (full carriage tilt backward). Measure the vertical distance from the surface to the lowest point on the vehicle, excluding wheels and tires. This is the underclearance.
   b. Test course 1b above. Drive the forklift with rated load from one horizontal surface up the ramp to the other horizontal surface in forward direction so that all wheels are on the horizontal surface. Descend the ramp in reverse.

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TEST METHOD NO. 5

UPRIGHT TILT - TILT SPEED - STATIC STABILITY

TILT CAVITATION - MAST CLEARANCE

1. TEST COURSE:

   a. Level, flat surface.

2. TEST APPARATUS:

   a. Clinometer.
   b. Stopwatch or electronic timer.
   c. Rigid pole of a length not less than the width of the forklift.
   d. Hydraulic pressure gage capable of reading below atmospheric pressure.
   e. Thermocouple.

3. TEST PROCEDURE:

   a. Upright tilt.

      (1) Place forklift without load on level, flat surface, using clinometer to insure forklift is level.
      (2) Raise forks above the surface and tilt as far forward as possible and record angle on clinometer located on front surfaces of outer mast channel.
      (3) Tilt forks as far rearward as possible and record angle of tilt on clinometer.

   b. Tilt speed and static stability.

      (1) Chain the rated load to the carriage and lift the load to maximum lift height.
      (2) Tilt the load to maximum forward tilt.
      (3) At governed engine speed tilt the load from maximum forward tilt, to maximum rearward tilt, and record the time required..
      (4) Repeat the procedure tilting from rearward to forward at low engine idle speed. Hinged frame steer forklifts shall be at the maximum steer angle. Suitable safety precautions should be observed.

   c. Tilt cylinder cavitation.

      (1) Install the pressure gage to read hydraulic pressure at the head end of the tilt cylinder. Install thermocouple to read oil temperature at the hydraulic tank.
      (2) After heating the oil to specified temperature, lift the rated load to maximum fork height.

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TEST METHOD NO. 5 (Continued)

(3) With the engine at idle speed, slowly tilt the load from full rearward to full forward and back agian. Tilt forward shall be done with control valve wide open 5 times and "feathered" 5 times. Observe and record any evidence of cavitation in the tilt circuit or loss of operator control.

d. Tire clearance.

(1) Position the rigid pole across the forks and anchor it as far rearward as possible on the horizontal fork surface.
(2) Operate the forks thoughout the full range of lift heights and tilt angles observing any interference between the pole and any part of the forklift.

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TEST METHOD NO. 6

DRIFT

1. TEST COURSE:

   a. Level, flat surface.

2. TEST APPARATUS:

   a. Temperature thermocouple and potentiometer or equivalent temperature measuring device.
   b. Clinometer.

3. TEST PROCEDURE:

   a. Install the thermocouple in the volumetric center of the hydraulic-system oil reservoir. Heat the hydraulic oil by exercising hydraulic functions until the oil temperature exceeds 120° F. Maintain oil temperature above 120° F for the duration of the test.
   b. Raise rated load to maximum lift height. Tilt mast to true vertical position measured with the clinometer. Mark reference point.
   c. Hold the load for 2 minutes and record the downward drift and rotational drift in that period.

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TEST METHOD NO. 7

LONGITUDINAL GRADEABILITY

1. TEST COURSE:

   a. The length of the test course shall be not less than three times the wheel base of the forklift, graded to 45 percent within ±1 percent grade. Grade resistance shall be sufficient to prevent observable wheel slip. Test course surface shall be unpaved.

2. TEST APPARTUS:

   a. Speed indicator.
   b. Exhaust system pressure measurement device.

3. TEST PROCEDURE:

   a. Install exhaust system pressure measurement device not more than 2 inches beyond the header outlet. Record exhaust pressure for each run.
   b. Drive the forklift with rated load forward up the slope, entering the course at a speed of not less than 2 mph, nor more than 4 mph. Continue through the course, maintaining a speed of not less than 2 mph. Make a total of three such passes up the slope.

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TEST METHOD NO. 8

DYNAMIC STABILITY - PARKING BRAKE - EMERGENCY BRAKE

1. TEST COURSE:

a. The length and width of the test course shall be not less than the forklift curb clearance circle diameter, graded to 30 percent slope, within ±1 percent grade. Test course surface shall be unpaved.

2. TEST APPARATUS:

a. Force measurement equipment.

3. TEST PROCEDURE:

a. Full circle performance.

(1) Drive the forklift with rated load centered on the forks onto the slope and perform a complete forward circle, at maximum steer angle, while on the slope.
(2) Repeat the procedure in 3a(1), steering in the opposite direction.
(3) Repeat 3a(1) and 3a(2), without load.

b. Slope descension, braking, and parking brake operation.

(1) Position the forklift with rated load at the top of the slope.
(2) Drive the forklift with rated load forward down the slope until a speed of not less than 5 mph is attained; then stop and hold the forklift with the service brakes for not less than 1 minute, with the transmission in forward gear range.
(3) Lower the load to ground level. Back away from the load until the forklift clears the load, and stop the truck with the service brake.
(4) Apply the parking brake. Measure and record the force necessary to apply the parking brake at the center of the handgrip.
(5) Turn the truck around so that it is facing forward up hill. Repeat 3b(4).

c. Emergency brake operation.

(1) Position the forklift with rated load at the top of and facing down the slope. Shut off the engine and allow the forklift to roll down the slope in neutral gear range, until a speed of 2 mph is attained. Apply the emergency brake provisions (parking or service brake as applicable) and bring the forklift to a stop. Release the brake, allow the forklift to begin to roll, and apply the brake again to stop the forklift, and repeat for a total of six dead engine stops.

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TEST METHOD NO. 9

STEERING

1. TEST COURSE:

   a. The test course shall be a dry, level surface in accordance with figures A-1 and A-2.

2. TEST APPARATUS:

   a. Speed indicator.
   b. For this test only, rated capacity load (see 6.9.2) may be substituted with a 4000 lb 24-inch load center load with height at the option of the contractor.

3. TEST PROCEDURE:

   a. Obstacle course No. 1 in accordance with figure A-1. Drive the forklift forward, without load, over the test course twice, once in each direction at a speed not less than 20 mph.
   b. Repeat the procedure in "a" with rated load in carry position, (full tilt backward) at a speed not less than 20 mph.
   c. Obstacle course No. 2 in accordance with figure A-2. Drive the forklift forward, without load, over the test course, at full-load governed speed in first gear. Repeat with load in carry position.

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TEST METHOD NO. 10

EMERGENCY STEERING

1. TEST COURSE:

   a. The test course shall be a dry, level surface in accordance with figure A-3.

2. TEST APPARATUS:

   a. Speed indicator.
   b. For this test only, rated capacity load (see 6.9.2) may be substituted with a 4000 lb 24-inch load center load with height at the option of the contractor.

3. TEST PROCEDURE:

   a. Drive the forklift with rated load a sufficient distance to attain a speed of not less than 20 mph. When the forklift reaches point A (figure A-3) at that speed, shut off the engine, shift the transmission lever to neutral and back into high gear. Slow the vehicle by use of the service brakes, if necessary. At point B, steer the forklift off the course, and come to a stop parallel to the course (point C). The forklift shall be under the control of the operator at all times and shall contact all of the markers on the 100 foot radius curve while negotiating the curve.

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TEST METHOD NO. 11

STEERING CONTROL

1. TEST APPARATUS:

   a. Force measurement equipment.
   b. Stopwatch.
   c. Linear measurement equipment.

2. TEST PROCEDURE:

   a. With the engine operating at low idle, determine the time, and the tangential force at the steering wheel rim, required to turn the wheel from stop to stop in both directions. Repeat 2 more times.
   b. Repeat 2a except that engine shall operate at maximum governed speed.
   c. Count the number of revolutions required to turn the wheel from stop to stop in each direction. Repeat 2 more times.
   d. Apply a minimum 300 pound force downward perpendicular to the rim of the steering wheel approximately midway between two spokes and hold for a minimum of 10 seconds.
   e. Repeat 2d except the force shall be applied upward.
   f. With engine at low idle, apply the maximum force attainable up to 150 pounds, to the rim of the steering wheel, and hold for a mimimum of 10 seconds.
   g. Repeat 2f in the opposite direction.
   h. Measure seat to steering wheel clearance and directional control to steering wheel clearance.

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TEST METHOD NO. 12

HIGHWAY AND RAIL TRANSPORTABILITY

1. TEST COURSE:

a. Paved roadway.
b. Railroad hump facility.

2. TEST APPARATUS:

a. Towing vehicle with pintle hook and speed indicator.
b. Suitable rail-hump equipment.

3. TEST PROCEDURE:

a. Highway transportablity (towing).

(1) Disengage the wheels from the transmission using one man provided with common tools. Record the time necessary to perform the operation.
(2) Attach the towbar and safety chains to the towing vehicle and pull the forklift for not less than 50 miles at 35 mph, ±5mph. Observe tracking ability and note any loss of control.
(3) At the conclusion of the 50-mile-towing, disconnect the forklift from the towing vehicle, reengage the driveline and operate the forklift to observe for damage.

b. Rail transportability (rail hump). The truck shall be blocked, braced, and tied down in accordance with procedures contained in section 6, figure 54 of Association of American Railroads (AAR) Rules Governing the Loading of Department of Defense Material on Open Top Cars. Hump the car, with the truck tied down at a speed of not less than 8 mph against five unloaded boxcars with their brakes set. The truck shall face in the direction of the five cars. Repeat the procedure at not less than 8 mph with the rear of the truck in the direction of the five cars. The truck may be reversed by either reloading it in the specified direction on the flatcar or by reversing the direction of the flatcar. Once the test has begun, there shall be no readjustment of load, nor any reconditioning of the blocks, chocking material, or tiedowns except to reverse direction of the truck. After each impact, the truck and tiedowns, shall be inspected to determine the amount of load shift, permanent deformation to components of the truck, or other evidence of possible failure.

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TEST METHOD NO. 13

OVERLOAD

1. TEST COURSE:

   a. The test course shall be a flat, level surface with provisions for anchoring the forklift to prevent tipping and for blocking the front axle to prevent overloading the tires.

2. TEST APPARATUS:

   a. Test loads or other apparatus to apply load to forks.

3. TEST PROCEDURE:

   a. Anchor the forklift to prevent tipping and block the front axle to prevent overloading the tires. Raise the forks to provide not less than 8 inches ground clearance to level forks.
   b. Apply three times the rated load on the forks at 24 inch load center distance, and hold for 5 minutes.
   c. Sideshift the forks fully to the right and apply two times the rated load on the forks at 24 inch load center distance, and hold for 5 minutes.
   d. Sideshift the forks fully to the left and repeat the procedure in "c".
   e. Remove the load and examine the forklift for damage.

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TEST METHOD NO. 14

TORQUE CONVERTER - PINTLE HOOK

1. TEST APPARATUS:

   a. Timing device.
   b. Thermocouple and potentiometer or equivalent temperature measuring device.
   c. Tachometer.
   d. Thermometer.

2. TEST PROCEDURE:

   a. Install thermocouple to measure oil temperature at the outlet from the converter.
   b. Thermostat shall not be blocked open.
   c. With the ambient temperature not less than 25° F, operate the forklift until the torque converter oil outlet temperature is a minimum of 157° F.
   d. Attach the forklift pintle hook to a fixed object to prevent forward motion of the forklift by a horizontal force on the pintle hook. Shift into the highest forward gear and stall the torque converter for not less than 30 seconds while the engine is operating at governed speed. The transmission disconnect or inching control shall not be used during this test.
   e. Record initial torque converter oil temperature and maximum oil temperature after 30 seconds stall, governed speed at stall, and ambient temperature.

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TEST METHOD NO. 15

INCHING/TRANSMISSION DISCONNECT

1. TEST APPARATUS:

   a. Thermocouples and potentiometers or equivalent temperature measuring device.
   b. Timing device.
   c. Linear measurement equipment.
   d. Pedal force gage.

2. TEST PROCEDURE:

   a. Install thermocouples to record temperatures listed below:

      (1) Engine outlet to top tank of the radiator.
      (2) Engine oil gallery.
      (3) Transmission sump oil to cooler.
      (4) Transmission oil from cooler.
      (5) Hydraulic tank.
      (6) Ambient air (thermometer may be used).

   b. Block open thermostat.
   c. Place truck on level surface with forks elevated approximately 1 foot from the surface.
   d. With direction control in forward position, depress accelerator to provide governed speed of engine and at the same time actuate the inching pedal to maintain a speed of 1/2 to 1-1/2 miles per hour, or actuate the transmission disconnet to stop the forklift, as applicable. Continue for not less than 20 minutes. Record temperatures at the beginning of the period and at 5 minute intervals thereafter. Speed shall be computed from the total distance traveled over the 20 minute period.
   e. With inching/transmission disconnect pedal fully depressed, transmission in forward low range, continually raise and lower rated load for a minimum of 5 minutes while engine is at governed speed. Record initial and final temperatures and pedal force required to depress pedal.

Note: Oil temperatures shall not exceed 30° F above ambient at the start of the tests in 2d and 2e.

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TEST METHOD NO. 16

SERVICE BRAKE

1. TEST COURSE:

   a. The test course shall be a dry, level, improved surface allowing no penetration, with a grade not exceeding ±1 percent.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Detonator, brake apparatus, with powder loads, or equivalent apparatus.
   c. Speed indicator.
   d. Pedal force gage.

3. TEST PROCEDURE:

   a. Stopping distance.

      (1) Attach the pedal force gage to the brake pedal in a manner which enables force to be applied to the face of the gage.
      (2) Connect the detonator to be operated by a switch directly conncted to the brake pedal so that it is actuated during the first 1/4 inch of pedal travel. The service brake, without transmission disconnect, shall be used for this test.
      (3) Place the rated load on the forks in carry position.
      (4) Drive the forklift with rated load over the test course. When the forklift reaches minimum 15 mph, apply the brakes. Measure and record the pedal force and the distance between the point of brake pedal application and the point of stoppage. Perform the test four times in forward gear; two times in each direction on the course. Observe whether the front wheels lock or any wheel loses contact with the test course. Measure the amount of fishtail.

   b. Brake system proof.

      (1) Apply 250 pound force to the gage, hold for 1 minute and release. Repeat this operation for a total of 10 times.

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TEST METHOD NO. 17

BRAKE FADE

1. TEST COURSE:

   a. The test course shall be a dry, improved surface allowing no penetration, with grades not exceeding ±1 percent.

2. TEST APPARATUS:

   a. Speed indicator.
   b. Decelerometer.

3. TEST PROCEDURE:

   a. Drive the forklift with rated load up to a minimum of 15 mph and apply the service brakes to produce a deceleration rate of 8 fps per second, ±1 fps per second, until the forklift has slowed to a minimum of 5 mph. Immediately accelerate the forklift at full throttle in high gear until the travel speed reaches a minimum of 15 mph. Immediately repeat the test for a total of 35 cycles. On completion of the 35th cycle, accelerate the forklift at full throttle in high gear to a minimum 15 mph and immediately conduct a service brake stopping test using procedure in test method No. 16. After the brakes have completely cooled, perform the test consisting of 35 snubs and one measured stop a second time. Make no adjustments or repairs between the two test runs.

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TEST METHOD NO. 18

WET BRAKE

1. TEST COURSE:

   a. Body of fresh water of sufficient depth to permit submersion of the forklift to a minimum 20 inch depth.

2. TEST PROCEDURE:

   a. Drive the forklift with rated load into the water to the minimum 20 inch depth. Release the service brakes and allow the forklift to stand for not less than 15 minutes. After the 15 minutes have elapsed, drive the forklift from the water and immediately conduct a service brake stopping test, using the procedure specified in 3a of test method No. 16. Service brakes shall not be applied after release in the water until the test application.

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TEST METHOD NO. 19

MAINTENANCE

1. TEST FACILITY:

   a. The test facility shall consist of a shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

   a. Common handtools.
   b. Special tools (if furnished by the contractor with the forklift).

3. TEST PROCEDURE:

   a. Examine the forklift for accessibility of major assemblies.
   b. Examine position of drains with respect to accessibility.
   c. Using the tools, demonstrate that the major assemblies are accessible for repair and maintenance.
   d. Demonstrate the accessibility of drains and path of discharge of lubricants and coolants by activating drainage controls.
   e. Perform and time each operation on the maintenance operations list (see 3.7).

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TEST METHOD NO. 20

PUMP INLET PRESSURE

1. TEST APPARATUS:

   a. Pressure gage.
   b. Hydraulic fittings (as required).
   c. Thermocouple.

2. TEST PROCEDURE:

   a. Install a pressure (vacuum) gage in the pump suction line immediately adjacent to the pump (or at the pump) and a thermocouple in the hydraulic tank.
   b. Heat the oil to specified temperature.
   c. Run the pump at the maximum speed it can encounter during vehicle operation and record the pump-inlet pressure, and temperature.
   d. Repeat for each pump installation on the truck.

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TEST METHOD NO. 21

HYDRAULIC FILTER AND FILTER ASSEMLBY

1. TEST APPARATUS:

   a. Hydraulic test stand may be used in place of on-vehicle test.
   b. Pressure gages.
   c. Temperature gages.
   d. Flowmeter (when test stand is used).
   e. Provisions for controlling oil temperature, if necessary.

2. TEST PROCEDURE:

   a. Pressurize the housing to not less than the minimum proof pressure (100 psi for return line filters, 30 psi for suction line filters, or 1-1/2 times maximum system pressure for pressure filters). Hold the pressure for not less than 60 seconds and examine for leaks.
   b. Install a new element in the filter housing. Establish maximum flow through the filter at an oil temperature of 150° F, ±5°F. Measure and record the pressure drop across the filter assembly.
   c. Plug the filter element chamber oil exit, or otherwise obstruct all flow through the filter element, leaving the outlet port open. Establish and maintain the oil temperature at 150° F, ±5° F. Increase the upstream pressure to 10 psi more than determined in "b" or to 2 psi for suction line filters. Observe the contamination indicator. Increase upstream pressure until the indicator enters the bypass area or until bypass flow exceeds 5 percent of the maximum system flow, whichever occurs first. Observe and record this pressure. (This is the bypass opening pressure.) Increase pressure until the maximum system flow is attained through the bypass valve. Observe and record this pressure.

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TEST METHOD NO. 22

HYDRAULIC SYSTEM CLEANLINESS

1. TEST APPARATUS:

   a. Liquid automatic particle counter.

2. TEST PROCEDURE:

   a. Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
   b. Run the engine at no load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in "d" below.
   c. When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in "d" below.
   d. The procedure for determining contamination shall be as follows:

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
      (2) Use any combination of on-the-forklift and off-the-forklift filtration for cleanup.
      (3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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TEST METHOD NO. 23

CYLINDER PROOF

1. TEST APPARATUS:

   a. Hydraulic test bench or the truck capable of developing two times system working pressure.
   b. Pressure gage.
   c. Thermocouple.

2. TEST PROCEDURE:

   a. Double acting cylinder.

      (1) Position and mechanically hold the piston at the approximate midpoint of the cylinder.
      (2) Fill both sides of the piston with oil at a minimum of 120° F.
      (3) With the head end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for a minimum of 60 seconds. Release pressure.
      (4) With the rod end port capped, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for a minimum of 60 seconds. Release pressure.

   b. Single acting cylinder.

      (1) Extend cylinder to maximum extension by filling with oil at a minimum of 120° F.
      (2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for a minimum of 60 seconds. Release pressure.

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TEST METHOD NO. 24

CONTAINER OPERATION

1. TEST FACILITY:

   a. The test facility shall consist of a flat, level area of sufficient size to permit operation of the forklift around the container, ramp, and trailer chassis.

2. TEST APPARATUS:

   a. Container ANSI MH5.1 and MH5.1.1, 20 feet long, 8 feet wide x 8 feet high.
   b. Ramp: Loading vehicle in accordance with MIL-R-52899.
   c. Container chassis minimum 55 inch height.

3. TEST PROCEDURE:

   a. Place the container on the chassis or platform. Place the end of the ramp into the container. Drive the forklift up the ramp and all the way into the container. Operate the forklift sideshift and raise forks to free lift height while in the container. Back the forklift out of the container and down the ramp to ground level. Check for interferences between the truck and container while entering he container, operating the container, and exiting the container. Check for visibility of the ramp curb in the area of the left rear wheel of the truck when backing out of the container.

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TEST METHOD NO. 25

SIDESHIFT, OSCILLATION, AND FORK SPREAD

1. TEST APPARATUS:

a. Clinometer.
b. Linear measurement equipment.
c. Stopwatch.

2. TEST PROCEDURE:

a. Clockwise oscillation.

(1) With the forklift on a level surface, place rated load on the forks and sideshift the load to extreme left side position. Place a clinometer on the load or carriage.
(2) Oscillate the load to extreme counterclockwise position and record the angle of oscillation relative to horizontal obtained.
(3) Oscillate the load to clockwise position and record the angle of oscillation relative to horizontal obtained.
(4) Repeat 2a(2) and 2a(3) at low engine idle and high engine idle a minimum of 3 times each and record time for each.

b. Counterclockwise oscillation.

(1) With the forklift on a level surface, place rated load on the forks and sideshift the load to extreme right side position. Place a clinometer on the load or carriage.
(2) Oscillate the load to extreme clockwise position and record the angle of oscillation relative to horizontal obtained.
(3) Oscillate the load to extreme counterclockwise position and record the angle of oscillation relative to horizontal obtained.
(4) Repeat 2b(2) and 2b(3) at low engine idle and at high engine idle a minimum of 3 times and record time for each.

c. Forks and fork spacing.

(1) With no load on the forks, move the forks as close together as possible and measure the distance between the inside edges of the forks. Move the forks as far apart as possible and measure the distance over the outside edges of the forks.
(2) Measure the fork length, fork width, fork thickness at the heel and at the tip, and the fork taper.

d. Sideshift.

(1) Spread the forks to a minimum 34 inches (outside to outside).

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TEST METHOD NO. 25 (Continued)

(2) Sideshift level forks to extreme left position. Note and measure any increase in truck width and whether there are any protusions beyond the sideshift carriage.
(3) Sideshift forks to extreme right position. Measure sideshift travel. Note and measure any increase in truck width and whether there are any protrusions beyond the sideshift carriage.
(4) Sideshift forks to extreme left position and measure travel.
(5) Place rated load on the forks. Sideshift to the extreme left position. Oscillate to extreme counterclockwise position.
(6) Oscillate the load to extreme clockwise position. Sideshift load to extreme left side position.
(7) With rated load on the forks in the horizontal position, operate the sideshift function for 100 full cycles at maximum speed. During the last 50 cycles, the sideshift control shall be released abruptly when the load is in approximately midposition, and the cycle shall be continued.

Notes: Sideshift cycles may be operated in 10 cycle intervals to allow sufficient cooling of the system.

One cycle is defined as sideshifting from extreme left to extreme right and return to extreme left.

All directions are determined relative to the operator sitting forward in the operator's seat.

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TEST METHOD NO. 26

RAIN

1. TEST APPARATUS:

   a. Timing device.
   b. Measuring device (U.S. Weather Bureau type preferred).
   c. Spray fixture.

2. TEST PROCEDURE:

   a. Subject the forklift to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the forklift at an angle of 45 degrees, ±5 degrees, with a force equivalent to a 30 mile per hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the forklift.
   b. Expose the top and left side of the forklift to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.
   c. Repeat "b" with the top and front exposed.
   d. Repeat "b" with top and right exposed.
   e. Repeat "b" with top and rear exposed.
   f. Stop the rainfall and allow the forklift to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid reservoir (except cooling system) and examine for presence of water.
   g. Check electrical components, gages, cab (if furnished), and the air cleaner for water entry and damage.

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TEST METHOD NO. 27

LOW-TEMPERATURE TEST (-25° F)

1. TEST APPARATUS:

   a. Cold chamber.
   b. Arctic clothing.
   c. Thermocouples with potentiometer(s) or other temperature measuring devices.
   d. Battery hydrometer.
   e. Wattmeter or ammeter.
   f. Voltmeter.

2. TEST PROCEDURE

   a. Service the forklift with applicable lubricants for -25° F operation including engine oil, transmission oil, drivetrain lubricants, hydraulic oil and antifreeze.
   b. Check the specific gravity of battery electrolyte to insure full charge.
   c. Subject the forklift to an ambient temperature not greater than -30° F for a period of 6 hours without solar radiation. During and after this period, examine the forklift for damage. Subject the forklift to an ambient temperature of -25° F, ±5° F, until the engine coolant, fuel, and lubricating oil temperatures are stabilized from thermocouples located as follows:

      (1) In the two center cells of each battery between the plates and at middepth of the electrolyte.
      (2) In the volumetric center of the hydraulic system oil reservoir.
      (3) In the engine lubricating oil gallery and in the oil sump.
      (4) In the engine coolant in the engine cooling jacket discharge pipe.
      (5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the truck, not less than 3 feet from each corner of the truck at engine crankshaft level.

   d. Maintain -25° F, ±5° F, ambient temperature throughout the entire test.
   e. One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.
   f. Substitute or supplementary batteries or other electrical cranking energy shall not be used.
   g. Glowplugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the forklift and normally used in starting the engine may be used.
   h. Start the forklift. Starting should be accomplished within 5 minutes of the initial attempt. Smooth engine running without unnatural or continued control manipulation, and without use of primers and priming fluids, should be obtained within 15 minutes after starting.

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TEST METHOD NO. 27 (Continued)

i. When started, run engine 15 minutes. Ater completion of the 15 minute operation, operate the engine as follows:

(1) 10 minutes idling.
(2) 10 minutes half throttle.
(3) 5 minutes full throttle.

Perform the 25 minute cycle three times.

j. Record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during warmup and cycling, including but not limited to the following:

(1) Battery.
(2) Hydraulic tank.
(3) Engine oil.
(4) Engine coolant.
(5) Chamber air.

k. During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels in all gears.
(2) Operate through two complete movement cycles all brakes, shafts and cylinders (as permitted by space limitations in chamber).
(3) Mount to and dismount from the operator's position on the forklift.

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TEST METHOD NO. 28

PERFORMANCE (FORDING)

1. TEST COURSE:

a. The test course shall be a dry land area and a body of water of sufficient depth to permit submersion of the forklift to minimum 20 inch depth.

2. TEST PROCEDURE:

a. Paint a horizontal white line on each side of the forklift at the 20 inch fording depth.
b. Fluid samples shall be taken from the lowest point of the following at the beginning of the test, and at the conclusion of test:

(1) Torque converter sump.
(2) Engine oil sump.
(3) Transmission.
(4) Transfer case.
(5) Fuel tank.
(6) Hydraulic reservoir.
(7) Each axle.

Samples shall be taken not more than 10 minutes after engine shut down and exercise of all functions on land for not less than 15 minutes. Samples shall be replaced with fresh lubricant. Sample size shall be not less than 200 cubic centimeters each. Samples shall be analyzed for volumetric water content by a means having an accuracy of within ± 1 percent.

c. Operate the forklift as follows:

(1) Drive the forklift into the water to the specified depth. Maneuver the forklift for a period of not less than 15 minutes in the water as follows:

(a) Drive the forklift forward in first gear at maximum speed for a period of 3 minutes. Stop the forklift by applying the service brakes.
(b) Drive the forklift backward in low reverse at maximum speed for a period of 3 minutes. Stop the forklift by applying the service brakes.
(c) During the 6 minute cycle in (a) and (b), the steering system shall be actuated from stop to stop.
(d) At the end of the 6 minute cycle and while at maximum water depth, operate all hydraulic cylinders, except steering cylinders, through one complete movement cycle. Prior to leaving the water, shut down the engine and restart it.

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TEST METHOD NO. 28 (Continued)

(e) Repeat (a) through (d) one time.

(2) At the end of the 15 minute period, drive the forklift onto dry land and repeat the maneuvering for a period of 15 minutes.

d. Repeat the procedures in 2c four additional times for a total of not less than 2-1/2 hours.

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TEST METHOD NO. 29

LOW TEMPERATURE WINTERIZATION (-50° F)

1. TEST APPARATUS:

a. Cold chamber.
b. Arctic clothing.
c. Thermocouples and potentiometer(s) or equivalent temperature measuring device.
d. Battery hydrometer.
e. Wattmeter.
f. Voltmeter.
g. Ammeter.
h. Tachometer.
i. Vaporizong apparatus.

2. TEST PROCEDURE:

a. Service the forklift with arctic fuel, lubricant and antifreeze, purging all other fluids from all systems.
b. Check the specific gravity of each battery. Specific gravity shall be not lower than 1.275 before beginning of test.
c. Locate thermocouples as follows and record temperatures at 10 minute intervals throughout the test (additional temperature measurements may also be taken for informational purposes):

(1) In the center of two center cells of each battery between the plates and at middepth of the electrolyte.
(2) In the volumetric center of the hydraulic oil reservoir.
(3) In the engine lubricating oil gallery.
(4) In the engine coolant in the engine cooling jacket discharge pipe.
(5) In the chamber, at the four corners of the forklifts at crankshaft level, 3 feet from each corner at a 45 degree angle with the sides of the forklift.
(6) In the operator's compartment, 6 inches above the floor, between foot treadles.

d. Subject the forklift to an ambient temperature of -50° F, $\pm$5° F, for a period of not less than 6 hours.
e. While the chamber temperature is falling, spray the closed radiator winterfront and air intake with water until a coat of ice averaging 1/4 inch thick has accumulated on both. One man wearing actic clothing, using no tools, shall then open and close both the winterfront and air intake.
f. Remove a vial of fuel from the fuel tank and observe if the cloud point of the fuel is reached as the temperature drops.

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TEST METHOD NO. 29 (Continued)

g. When all measured temperatures stabilize at -50° F (but not before the 6-hour storage period has elapsed), initiate the starting procedure. Maintain -50° F, ±5° F, air temperature in the chamber throughout the test.

h. The starting procedure shall include a prestart warm-up period not to exceed 60 minutes. During this period, the engine coolant heater shall be operated, with its exhaust used to heat the battery box. The cab heater shall also be operated with its flow diverted to the batteries. Determine and record the starting and running power of each heater and its accessories. Record the time required for fuel ignition in the heaters.

i. After the battery electrolyte temperature reaches 50° F, but before the 60 minute warm-up period elapses, start the engine. Substitute batteries or external cranking power shall not be used. Glowplugs or other accessories working off the cranking batteries and priming systems which are permanently mounted on the forklift and normally used in starting the engine may be used.

j. Run the engine without using the engine coolant heater. Shut off all other cold-start aids within 30 minutes of engine start.

k. After the engine has started, divert cab heater air to the cab. Observe the operation of the defrosters in the first hour after sarting the engine with an operator in the cab or with the vaporizing apparatus used to duplicate moisture introduced into the air by an operator.

l. Operate the engine until the lubricating oil temperature and coolant temperature have stabilized.

m. Run the engine for an additional hour after these temperatures have stabilized, periodically cycling the engine between idling and full load governed speed. During this period, an operator wearing arctic clothing shall:

(1) Mount to and dismount from the operator's position on the forklift.
(2) Engage and disengage the power train to all wheels.
(3) Operate through two complete cycles all brakes, shafts, cylinders, and hydraulic components (as space permits).
(4) Operate windshield wipers.
(5) Operate the cab doors, and engine housing doors by opening, closing, unlocking, or latching as applicable.

n. Upon completion of these operations, shut down the engine. While the forklift is still in the chamber at -50° F, the operator, wearing arctic mittens, shall perform each of the following maintenance checks and operations:

(1) Restart the engine using outside power through the slave receptacle. Disconnect the cable to outside power and then shut down the engine.
(2) Remove and reinstall filter on each heater fuel pump.

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TEST METHOD NO. 29 (Continued)

(3) Remove and reinstall filter element in fuel filter in each heater fuel line.
(4) Check oil level on all dipsticks.
(5) Check battery electrolyte levels.
(6) Remove and reinstall all batteries.
(7) Drain approximately 1 gallon of coolant from the radiator, check radiator coolant level and add the drained fluid.

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TEST METHOD NO. 30

HIGH SPEED

1. TEST COURSE:

   a. The test course shall be a compacted surface of sufficient size to allow continuous travel at wide open throttle in highest gear range.

2. TEST APPARATUS:

   a. Speed indicator.
   b. Equivalent temperature measuring device.

3. TEST PROCEDURE:

   a. Block thermostat in the open position.
   b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter and engine oil gallery.
   c. Record all temperatures icluding ambient before starting test.
   d. Drive the forklift over the test course for 2 hours at wide open throttle in highest forward gear range. Stop the forklift and record all temperatures immediately, including ambient and universal joint temperatures. Check for universal joint temperature using temperature marker.
   e. In not less than 5 minutes after stopping, drive the forklift for an additional 2 hours as in "d", and record all temperatures immediately upon stopping, including ambient and universal joint temperatures.

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TEST METHOD NO. 31

PERFORMANCE (DURABILITY COURSE)

1. TEST COURSE:

   a. A test course shall be constructed in accordance with figure A-4.

2. TEST APPARATUS:

   a. Themocouples and potentiometer(s) or equivalent temperature measuring device.

3. TEST PROCEDURE:

   a. Block thermostat in the open position.
   b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter, and the engine oil gallery. One day per week, measure the temperatures, including ambient, at successive hourly intervals for a period of 6 hours.
   c. Drive the forklift, with lights on, with rated load in carry position (see 6.9.3). Average speed over the course shall be between 3.5 mph and 4.0 mph. For this test, the engine governor setting may be changed or accelerator stop may be installed to maintain speed throughout the test. The operator shall not vary the accelerator position while negotiating the test course between points C and D.
   d. One cycle is completed each time the forklift travels from loop A to loop B and returns to loop a. Turn around in loop A shall consist of right hand turns and turn around in loop B shall consist of left hand turns.

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TEST METHOD NO. 32

PERFORMANCE (CONTAINER/TRAILER LOADING)

1. TEST COURSE:

   a. A test course shall be constructed in accordance with figure A-5.

2. TEST APPARATUS:

   a. Thermocouple and potentiometer or equivalent temperature measuring device.
   b. Palletized loads, 16 each, mounted on 40 by 48 inch pallets, load enclosed in a box between 41 and 43 inches overall height including pallet. The gross weight of the pallets shall be: Six 4000-pound, five 2500-pound, and five 1000-pound, ±100 pounds each. The pallets shall be suitable for stacking up to three high, and shall provide adequate clearance underneath to be engaged by the forks, but not in excess of 6 inches.
   c. Container, ANSI MH5.1 and MH5.1.1, 20 feet long, mounted on a trailer chassis with auxiliary support for the fifth wheel.
   d. Container, ANSI MH5.1 and MH5.1.1, 20 feet long, on the ground.
   e. Trailer, highway type, flatbed large enough to carry all 16 pallets, not stacked.
   f. Ramp: Loading vehicle in accordance with MIL-R-52899.
   g. Fence or barricade, approximately 4 feet high by 60 feet long.

3. TEST PROCEDURE:

   a. Block thermostat in the open position.
   b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter, and the engine oil gallery. One day per week measure the temperatures, including ambient, at successive hourly intervals for a period of 6 hours.
   c. Handle all pallets from the 40 inch end. Start the test with all pallets in the trailer mounted container (four rows of two 2-high stacks).
   d. Move all pallets to the pallet stack area, stacking 3 high (five 3-high stacks and one not stacked). When stacking, the forklift (except forks) shall remain between the barricade and the marked boundary. The forklift tires shall not cross the marked boundary during any portion of the test.
   e. Move all pallets to the flatbed trailer, loading from both sides of the trailer and not stacking (two parallel rows of 8 pallets). The first pallet moved to the trailer shall not be the last one that was placed in the pallet stack.

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TEST METHOD NO. 32 (Continued)

f. Move all pallets to the container on the ground, stacking as in "c". The first pallet moved to the container shall not be the last one that was placed on the trailer.
g. Move all pallets to the trailer mounted container, stacking as in "c". The first pallet moved to the trailer mounted container shall not be the last one that was placed in the container on the ground.
h. Steps "d" through "g" constitutes one cycle. A cycle is completed each time the sixteenth pallet is loaded into the trailer mounted container.

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TEST METHOD NO. 33

CONTROLS

1. TEST APPARATUS:

   a. Thermocouples and potentiometer or equivalent temperature measuring device.
   b. Force measurement equipment.
   c. Linear measurement equipment.

2. TEST PROCEDURE:

   a. Install the thermocouple in the hydraulic reservoir.
   b. With an oil temperature of 140° F to 180° F and the engine operating at no load governed speed, measure the force and distance rquired to shift each hydraulic control valve at the handgrip to each position.
   c. Apply 100 pounds of force to each valve lever in each shifting direction and maintain the force for not less than 15 seconds.
   d. Apply 100 pounds of force to each transmission control lever in each direction and maintain the force for not less than 15 seconds.

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TEST METHOD NO. 34

HYDRAULIC FITTINGS

1. TEST APPARATUS:

   a. Common handtools.

2. TEST PROCEDURE:

   a. Remove one hydraulic hose assembly that has permanent-type fittings.
   b. Replace the hose assembly with an equivalent field attachable fitting only to assure interchangeability.
   c. Repeat "a" and "b" above until each hose assembly with permanent-type fittings is replaced with a hose assembly with reusable fittings.
   d. After every hose assembly with permanent-type fittings has been replaced with a hose assembly with reusable fittings, reinstall the original hose assemblies on the truck.

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TEST METHOD NO. 35

REAR AXLE OSCILLATION

1. TEST APPARATUS:

   a. Clinometer.

2. TEST PROCEDURE:

   a. Raise the forklift, keeping the forklift horizontal, to sufficient height to allow the rear axle to oscillate from stop to stop.
   b. Attach the clinometer to the rear axle.
   c. Oscillate the rear axle fully clockwise. Measure and record the angle from the horizontal.
   d. Oscillate the rear axle fully counterclockwise. Measure and record the angle from the horizontal.

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TEST METHOD NO. 36

NUCLEAR-BIOLOGICAL-CHEMCIAL

1. TEST COURSE:

   a. The test course shall be a level, improved surface, laid out in accordance with course B of figure A-4.

2. TEST APPARATUS:

   a. Palletized loads (three each rated load), with maximum 6-inch high fork pockets.
   b. Two platforms (one 8 feet 6 inches high, the other 8 feet high).
   c. The complete protective ensemble.

3. TEST PROCEDURE:

   a. While wearing the complete protective ensemble, each and every control (i.e., levers, buttons, switches, pedals, knobs and handles) shall be actuated by the operator. Record any incident of interference caused by the ensemble that is detrimental to the operation of the forklift.
   b. On course B, position the platforms at points 3 and 2 with the inside top of the platform at the 50-foot radius.
   c. Position the loads at points 1, 4, and 5 on course B with the inside edge of the load at 50-foot radius.
   d. The operator shall not be given any assistance in engaging any loads during this test.
   e. Start at point 1, pick up the load, and move to point 3. Place the load on the platform at point 3 and proceed to point 5. Pick up the load at point 5 and move it to point 2. Place the load on the platform at point 4 and move it to point 1. Continue the cycle, moving from point 1 to point 3, from point 3 to 5, etc. The forklift shall maneuver within the area of the course during all operations.
   f. Record any detrimental effects the ensemble has on the operational capabilities of the forklift.

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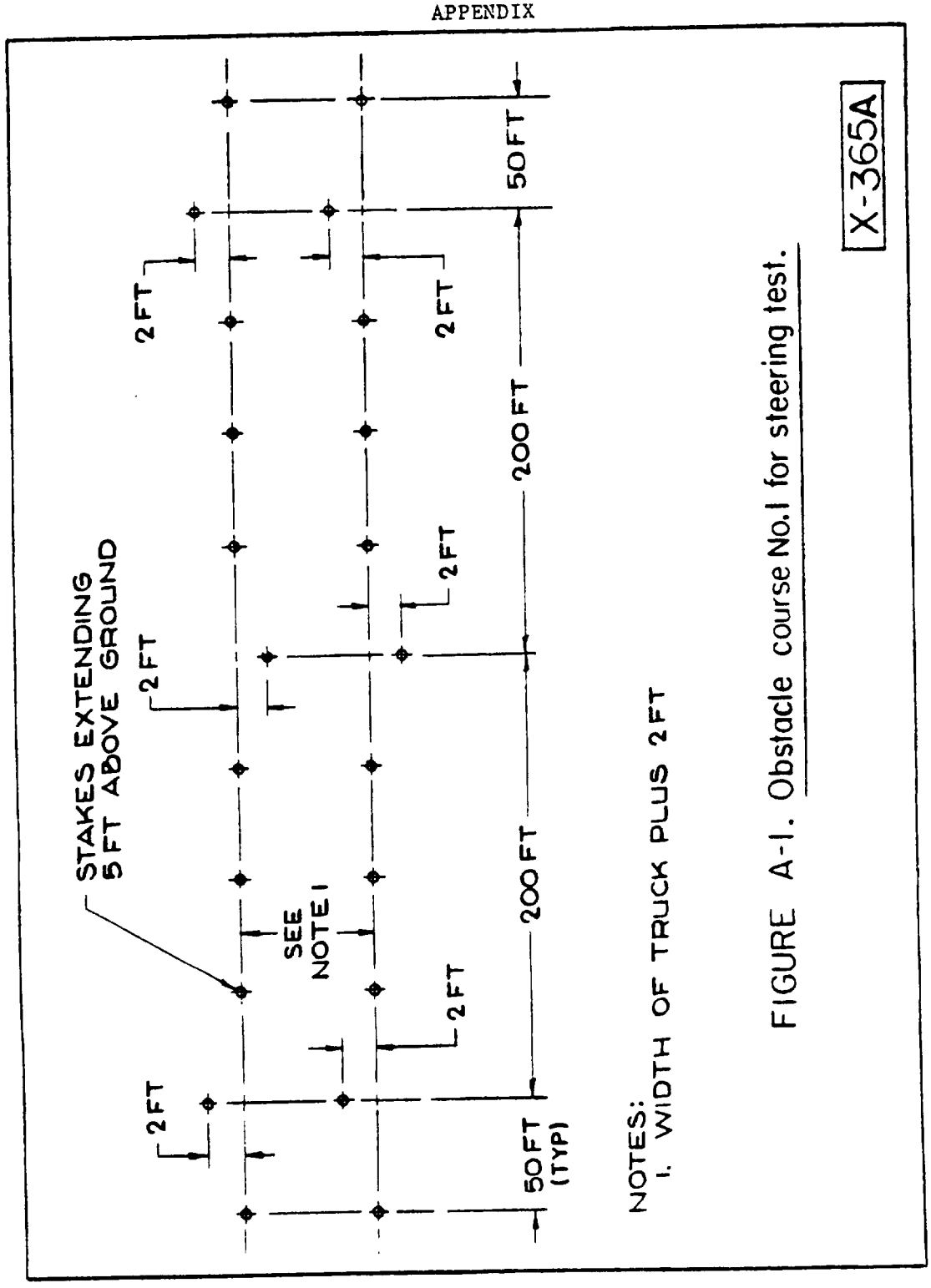
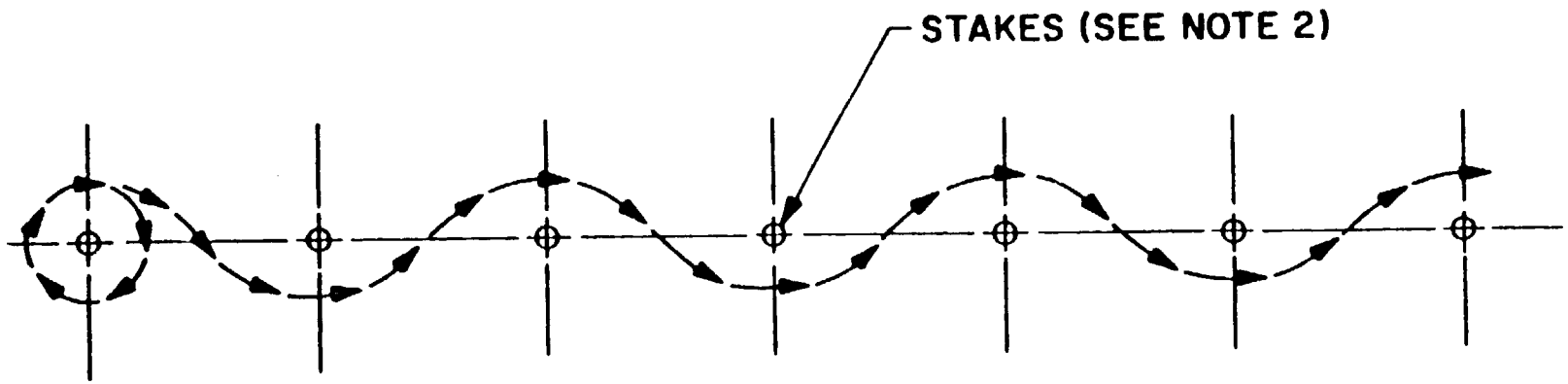


FIGURE A-1. Obstacle course No.1 for steering test.

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(SEE NOTE 1)

NOTES:

1. THE SPACE BETWEEN THE STAKES SHALL BE EQUAL TO TWO TIMES THE CURB CLEARANCE CIRCLE DIAMETER.
2. STAKES SHALL BE HIGH ENOUGH TO BE EASILY SEEN BY THE OPERATOR WHILE GOING THROUGH THE TEST COURSE.

FIGURE A-2

FIGURE A2. Obstacle course steering test.

X-2983E

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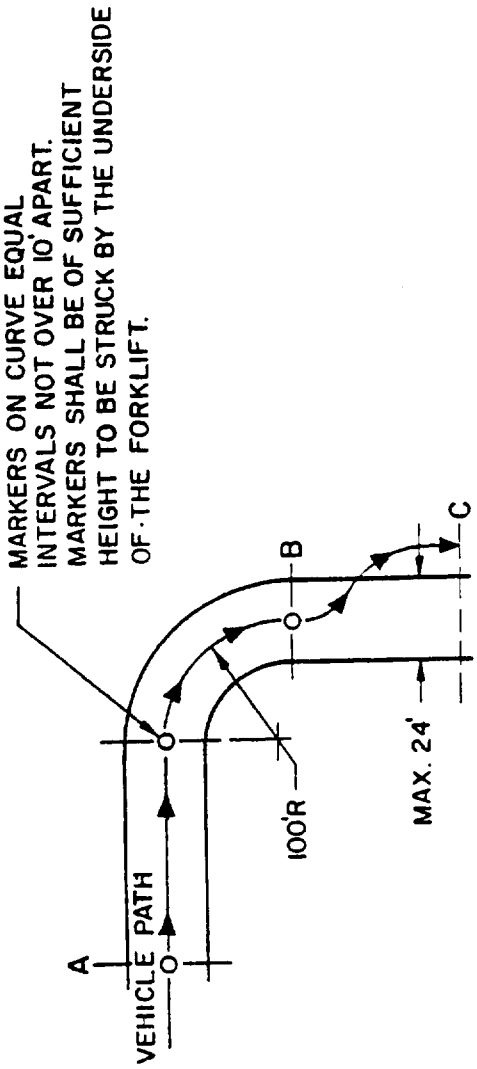
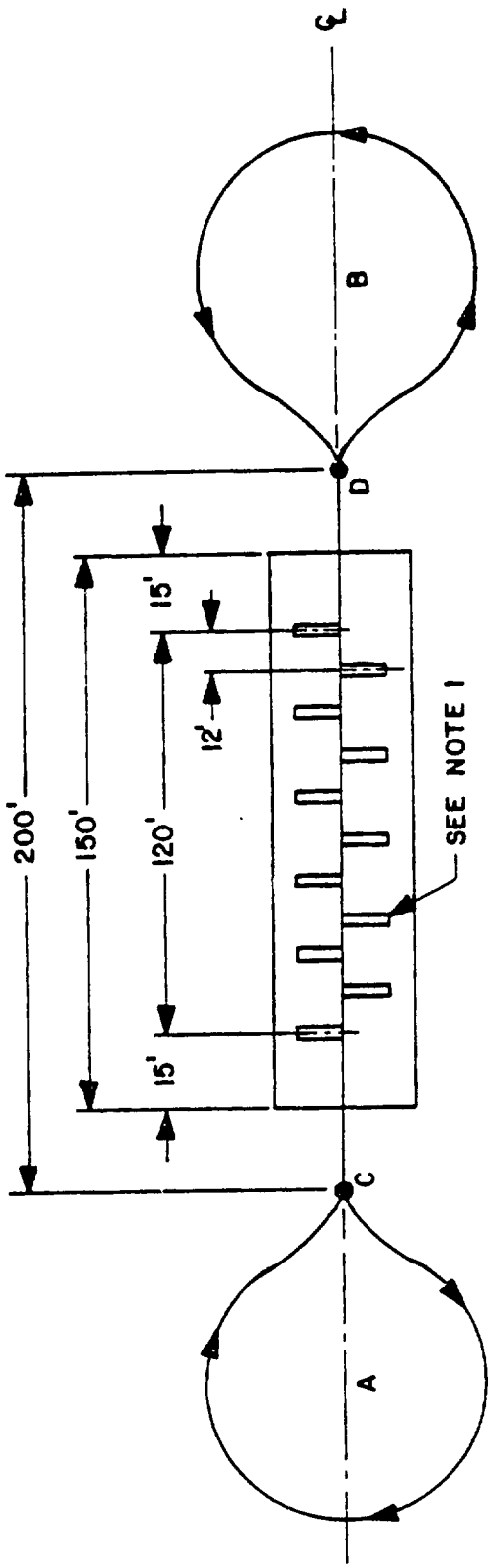


FIGURE A-3 TEST COURSE FOR EMERGENCY STEERING TEST X-2984

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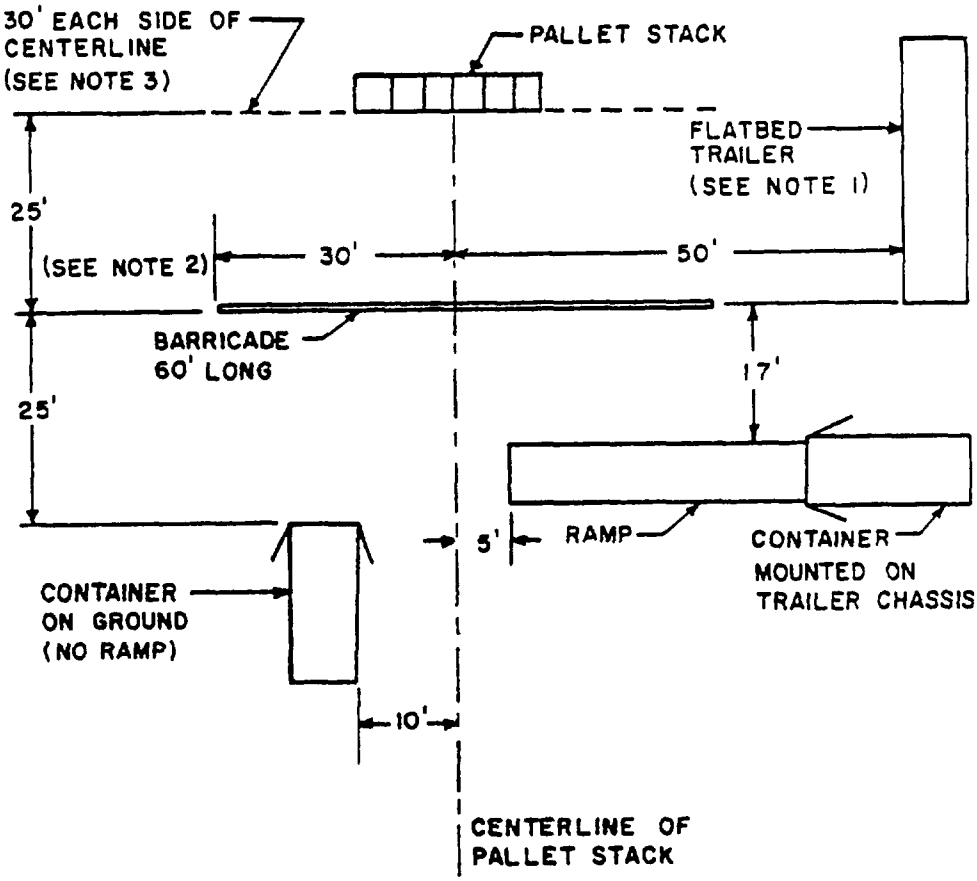
NOTES:

1. ELEVEN CROSSTIES, 6 FT. LONG, 12 FT. APART, ANCHORED TO PAVEMENT TO PREVENT MOVEMENT. CROSSTIES SHALL BE MINIMUM 2-1/2 INCHES HIGH X 4 INCHES WIDE.
2. SPEED SHALL NOT VARY BETWEEN POINTS C AND D.
3. TEST COURSE SHALL NOT BE PAVED EXCEPT FOR CROSSTIE AREA.

X-3351

FIGURE A-4. Typical durability course.

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NOTES:

1. ONE END OF FLATBED TRAILER ELEVATED TO PUT BED AT 8° ANGLE WITH THE HORIZONTAL.
2. ALL TRAVEL AROUND BARRICADE TO BE BETWEEN BARRICADE AND FLATBED TRAILER, EXCEPT WHEN APPROACHING RAMP TO ENTER TRAILER MOUNTED CONTAINER.
3. FORKLIFT SHALL NOT CROSS MARKED BOUNDARY.
4. PLATFORMS OR BOXES MAY BE USED TO DUPLICATE ACTUAL TRAILERS OR CONTAINERS.
5. TEST COURSE SHALL NOT BE PAVED.

FIGURE A-5. Container/trailer loading course.

X-3019A

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Load Center	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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